PRIμS:

Paleo-astroparticles Reconstructed with the Interactions of MUons in Stone

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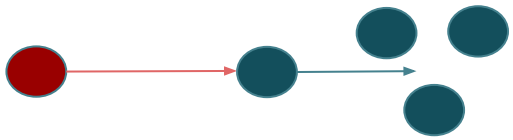
INFN sezione di Milano
LPC - Université Clermont-Auvergne

Presidenza INFN
19/11/2024

Mineral detection of astroparticles: the base idea

One of the many ways Geophysics has been connected to astroparticle physics over the years

Based on a very simple energy deposit idea: **Scattering** + **Track formation**



An energetic astroparticle - WIMP, neutrino, Muons from cosmic ray shower - **hits an atom in the crystal** lattice of a mineral. The atom recoils and **damages the crystalline structure** leaving a defect called ion track -or just **track**

Same formation **concept as fission tracks** - in that case the recoiling particle is the fission product

VOLUME 56, NUMBER 12 PHYSICAL REVIEW LETTERS 24 MARCH 1986

Search for Supermassive Magnetic Monopoles Using Mica Crystals

P. B. Price and M. H. Salamon

VOLUME 74, NUMBER 21 PHYSICAL REVIEW LETTERS 22 MAY 1995

Limits on Dark Matter Using Ancient Mica

D. P. Snowden-Ifft,* E. S. Freeman, and P. B. Price*
Physics Department, University of California at Berkeley, Berkeley, California 94720
(Received 20 September 1994)

The combination of the track etching method and atomic force microscopy allows us to search for weakly interacting massive particles (WIMPs) in our Galaxy. A survey of 80 720 μm^2 of 0.5 Gyr old muscovite mica found no evidence of WIMP-recoil tracks. This enables us to set limits on WIMPs which are about an order of magnitude weaker than the best spin-dependent WIMP limits. Unlike other detectors, however, the mica method is, at present, not background limited. We argue that a background may not appear until we have pushed our current limits down by several orders of magnitude.

PACS numbers: 95.35.+d, 14.80.Ly, 29.40.Ym, 61.72.Ff

The idea of looking for tracks caused by cosmic rays or more exotic particles has been floating around for 50 years

with interest from scientists such as Occhialini

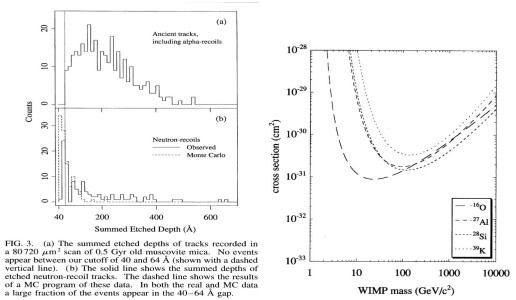
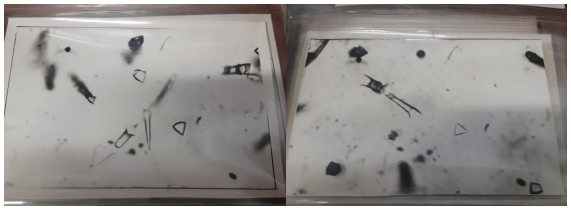


FIG. 3. (a) The summed etched depths of tracks recorded in a 80 720 μm^2 scan of 0.5 Gyr old muscovite mica. No events appear between our cutoff of 40 and 64 Å (shown with a dashed vertical line). (b) The solid line shows the summed depths of etched neutron-recoil tracks. The dashed line shows the results of a MC program of these data. In both the real and MC data a large fraction of the events appear in the 40–64 Å gap.



Set aside because of limited modeling, **readout** and **computational capabilities**

State of the art:

The concept was revived at the end of the 2010s and got the name **paleodetectors**

PHYSICAL REVIEW D **103,** 123016 (2021)

Measuring solar neutrinos over gigayear timescales with paleo detectors

Natalia Tapia-Arellano* and Shunsaku Horiuchi†
Center for Neutrino Physics, Department of Physics, Virginia Tech, Blacksburg, Virginia 24061, USA

PHYSICAL REVIEW LETTERS **125,** 231802 (2020)

Featured in Physics

Measuring Changes in the Atmospheric Neutrino Rate over Gigayear Timescales

Johnathon R. Jordan,[1,*] Sebastian Baum,[2,3,†] Patrick Stengel,[3,‡] Alfredo Ferrari,[4] Maria Cristina Morone,[5,6] Paola Sala,[7] and Joshua Spitz[1,§]
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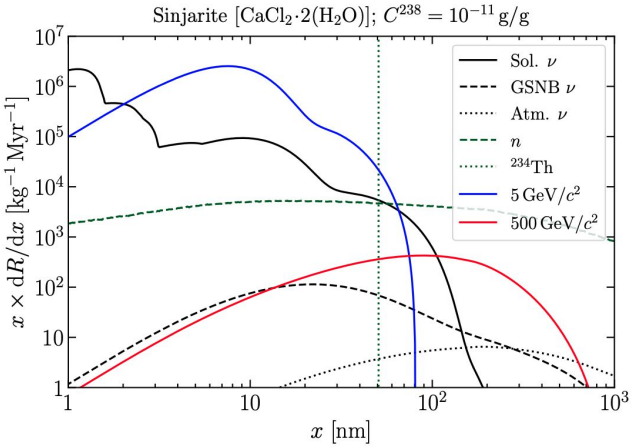
PHYSICAL REVIEW D **99,** 043541 (2019)

Digging for dark matter: Spectral analysis and discovery potential of paleo-detectors

Thomas D. P. Edwards,[1,*] Bradley J. Kavanagh,[1,†] Christoph Weniger,[1,‡] Sebastian Baum,[2,3,§] Andrzej K. Drukier,[2,‖] Katherine Freese,[2,3,4,¶] Maciej Górski,[5,*#] and Patrick Stengel[2,††]
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(Received 11 December 2018; published 27 February 2019)

The long exposure advantage: Paleodetectors capture and integrate exotic particle interactions **over geological timescales**, offering a high exposure detection scenario.



First goals:

- Model the **expected track length spectrum** from exotic messengers
- Establish feasibility and **identify** suitable mineral samples.

Tracks from secondary cosmic rays - mostly muons - in these studies are **the background:** the proposed samples are shielded deep underground.

Paleodetectors for cosmic rays

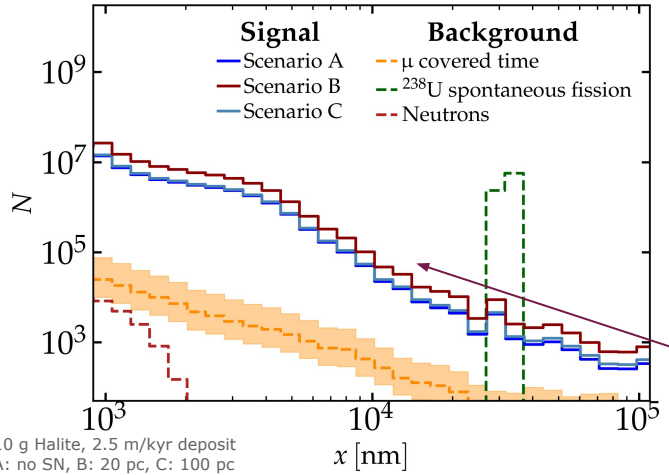
Beyond Background: Muon tracks aren't just noise for exotic particle studies they **hold valuable data about the past cosmic ray flux** in our galaxy and beyond!

The Historical Memory: we are interested in minerals that have been exposed to the CR flux for a period of time and then shielded by some material, keeping their accumulated historic signal.

Our example case study: the **Messinian salinity crisis**

A period ~6 Myr ago with desiccated Mediterranean. The seabed was covered in evaporites -mainly **Halite** Deposits were flooded and shielded after ~0.5 Myr





10 g Halite, 2.5 m/kyr deposit
A: no SN, B: 20 pc, C: 100 pc

Our simulations show that **tracks from secondary muons** are **background-less** except for the Uranium fission peak. With enough material there is possibility of **discrimination between different CR flux** scenarios.

The paleodetector community

Mineral Detection of Neutrinos and Dark Matter. A Whitepaper

Sebastian Baum,[1] Patrick Stengel,[2] Natsue Abe,[3] Javier F. Acevedo,[4] Gabriela R. Araujo,[5,a] Yoshihiro Asahara,[6] Frank Avignone,[7] Levente Balogh,[8] Laura Baudis,[5] Yilda Boukhtouchen,[9] Joseph Bramante,[9,10] Pieter Alexander Breur,[4] Lorenzo Caccianiga,[11] Francesco Capozzi,[12] Juan I. Collar,[13] Reza Ebadi,[14,15] Thomas Edwards,[16] Klaus Eitel,[17] Alexey Elykov,[17] Rodney C. Ewing,[18] Katherine Freese,[19,20] Audrey Fung,[9] Claudio Galelli,[21] Ulrich A. Glasmacher,[22] Arianna Gleason,[4] Noriko Hasebe,[23] Shigenobu Hirose,[24] Shunsaku Horiuchi,[25,26] Yasushi Hoshino,[27] Patrick Huber,[25,a] Yuki Ido,[28] Yohei Igami,[29] Yoshitaka Itow,[30] Takenori Kato,[30] Bradley J. Kavanagh,[31] Yoji Kawamura,[24] Shingo Kazama,[32] Christopher J. Kenney,[4] Ben Kilminster,[5] Yui Kouketsu,[6] Yukiko Kozaka,[33] Noah A. Kurinsky,[4,34] Matthew Leybourne,[9] Thalles Lucas,[9] William F. McDonough,[35,36,37] Mason C. Marshall,[15,38] Jose Maria Mateos,[39] Anubhav Mathur,[16] Katsuyoshi Michibayashi,[6] Sharlotte Mkhonto,[9] Kohta Murase,[40] Tatsuhiro Naka,[28] Kenji Oguni,[24] Surjeet Rajendran,[16] Hitoshi Sakane,[41] Paola Sala,[11] Kate Scholberg,[42] Ingrida Semenec,[9] Takuya Shiraishi,[28] Joshua Spitz,[43] Kai Sun,[44] Katsuhiko Suzuki,[45] Erwin H. Tanin,[16] Aaron Vincent,[9] Nikita Vladimirov,[46] Ronald L. Walsworth,[14,15,38] and Hiroko Watanabe[35]

Mineral Detection of Neutrinos and Dark Matter 2024 Proceedings

Sebastian Baum,[1] Patrick Huber,[2] Patrick Stengel,[3] Natsue Abe,[4] Daniel G. Ang,[5] Lorenzo Apollonio,[6] Gabriela R. Araujo,[7] Levente Balogh,[8] Pranshu Bhaumik[9] Yilda Boukhtouchen,[10] Joseph Bramante,[10] Lorenzo Caccianiga,[11] Andrew Calabrese-Day,[12] Qing Chang,[13] Juan I. Collar,[14] Reza Ebadi,[5] Alexey Elykov,[15] Katherine Freese,[16,17,18] Audrey Fung,[14] Claudio Galelli,[19] Arianna E. Gleason,[20] Mariano Guerrero Perez,[9] Janina Hakenmüller,[21] Takeshi Hanyu,[13] Noriko Hasebe,[22] Shigenobu Hirose,[4] Shunsaku Horiuchi,[2] Yasushi Hoshino,[23] Yuki Ido,[24] Vsevolod Ivanov,[9,25,26] Takashi Kamiyama,[27] Takenori Kato,[28] Yoji Kawamura,[4] Chris Kelso,[29] Giti A. Khodaparast,[25] Emilie M. LaVoie-Ingram,[29] Matthew Leybourne,[30] Xingxin Liu,[5] Thalles Lucas,[8] Brenden A. Magill[25] Federico M. Mariani,[6] Sharlotte Mkhonto,[30] Hans Pieter Mumm,[31] Kohta Murase,[32] Tatsuhiro Naka,[33] Kenji Oguni,[4] Kathryn Ream,[12] Kate Scholberg,[21] Maximilian Shen,[5] Joshua Spitz,[12] Katsuhiko Suzuki,[34] Alexander Takla,[12] Jiashen Tang,[5] Natalia Tapia-Arellano,[35] Pieter Vermeesch,[36] Aaron C. Vincent,[10] Nikita Vladimirov,[37] Ronald Walsworth,[5] David Waters,[38] Greg Wurtz,[29] Seiko Yamasaki[39] and Xianyi Zhang[40]

The community now spans tens of group, many in the US and Japan.

Italy with the **Milano** group is the **leader for the CR** interest group.

The community produced an 80+ page **"Whitepaper"** in 2023 and 2 general **workshops** (Washington 24)

Volume 77, Issue 11
1 November 2024



< Previous Article Next Article >

Search amps up for signatures of cosmic particles in ancient minerals FREE

Advances in imaging and data handling open new possibilities.

Toni Feder

Check for updates

Physics Today **77** (11), 20–22 (2024);
https://doi.org/10.1063/pt.xutj.ibib

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What if a mineral could reveal what it's seen over the millions—or billions—of years it's been sitting deep within Earth? An interdisciplinary global network of scientists is reviving efforts to unlock the secrets that minerals hold. The top goals are to identify dark matter, learn more about neutrinos, and use both as cosmological probes.

The **growth "wave"** of the technique is reflected in the more frequent appearances in conferences, mentions in papers and interest of the public: **Feature article** in the November APS Physics Today

The paleodetector community

Investigating the origins of UHECRs using the Pierre Auger Observatory and paleodetectors: Claudio Galelli, **PhD Thesis**

27th European Cosmic Ray Symposium, Nijmegen, the Netherlands, 25-29/07/22: Claudio Galelli et al., **Poster: "Cosmic ray signatures in Paleo-Detectors to investigate the past activity of our Galaxy".** https://indico.nikhef.nl/event/2110/contributions/14008/

2022 Astroparticle Symposium, Saclay, France, 21-25/11/22: Claudio Galelli, **Invited talk: "Digging for astroparticles: the paleodetectors technique".** https://indico.ijclab.in2p3.fr/event/8374/contributions/28070/

Meudon, France, 16/11/2023, Seminar: "Rock of the Ages: techniques and challenges for paleodetectors".

Clermont-Ferrand, France, 05/04/2024, Seminar: "Catching cosmic messengers with minerals: techniques and challenges for paleodetectors"

38th International Cosmic Ray Conference, Nagoya, Japan, 07/23: Lorenzo Caccianiga, Claudio Galelli, Lorenzo Apollonio, Paolo Magnani **Poster: "The use of paleodetectors to investigate cosmic-ray fluxes throughout the history of Earth"**

8th Heidelberg symposium on high-energy gamma-ray astronomy, Milano, Italy, 09/24: Lorenzo Apollonio, Claudio Galelli, Lorenzo Caccianiga **Poster: "Do minerals know about cosmic rays?"**

Sedimentary rocks from Mediterranean drought in the Messinian age as a probe of the past cosmic ray flux

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INFN - Sezione di Milano Via Celoria 16 20133 Milan, Italy

Lorenzo Apollonio,† Federico Maria Mariani, and Paolo Magnani
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Alessandro Veutro
Università di Roma La Sapienza, I-00185 Roma, Italy
§

(Dated: November 7, 2024)

We propose the use of natural minerals as detectors to study the past flux of cosmic rays. This novel application of the *paleo-detector* technique requires a specific approach as it needs samples that have been exposed to secondary cosmic rays for a well defined period of time. We suggest here the use of the evaporites formed during the desiccation of the Mediterranean sea ∼6 Myr ago. These minerals have been created and exposed to the air or under a shallow water basin for ∼500 kyr before being quickly submerged again by a km-scale overburden of water. We show that, by looking at the damages left in the minerals by muons in cosmic ray showers, we could detect differences in the primary cosmic ray flux during that period, as the ones expected from nearby supernova explosions, below the percent-level. We show also that little to no background from radioactive contamination and other astroparticles is expected for this kind of analysis.

Published on PRD, 21/11/24
Selected for APS and **INFN press release**

Perfect time to capitalize on the theoretical work and interest in literature

PRIμS: time for action

Solid science case:

Our simulated case shows that there is a **solid theoretical base** with **very interesting results** that is **relatively easy to transfer** to experiment: it is time to crack open the rock!

Perfect timing:

The wave of interest in paleodetectors offers the **ideal moment to validate techniques** and lead the field. Other groups in the community are preparing to move to action with controlled experiments on shorter tracks

Relevance and innovation:

Testing cosmic-ray tracks in ancient minerals will set a **global benchmark** for paleodetector experiments, and will position PRIμS as a cornerstone for future studies in high-energy and cosmic ray astrophysics.

Scientific impact:

- **First detection of a track from an astroparticle source**
- Give insights into the historical flux of cosmic rays, with potential applications for dark matter and neutrino studies.
- Establish essential data and methodological standards and **pioneer the emerging field of paleo-astroparticle physics.**

PRIμS: milestones

Gold Milestone	Measure and compare the **spectrum of track lengths** observed in a mineral using optical microscopy with simulations, including any necessary adjustments to the simulations.	First set standard on track length spectra for the paleodetector community. Base for all the future experimental studies in the field.
Platinum Milestone	Definitively detect and **identify ion tracks** resulting from interactions with secondary CRs.	First scientific experimental result of the community. Birth of paleo-astroparticle physics.
Diamond Milestone	**Measure the variation** in track production rates across different geological samples as a proxy for cosmic ray flux variations.	One of the largest advancements in cosmic ray physics in the last decade.

Sample selection and retrieval

The selection of the **type** and **history** of the sample to use is crucial

Ideal sample: Halite from the MSC

Preserved information on the flux of cosmic rays during the MSC desiccation event with defined exposure time

- Samples already recovered from prior campaigns
- Well-studied history due to geological interest
- **Solid track spectrum modeling** from our study
- Reference material with different history is available (e.g. Death Valley)





Alternative sample: Volcanic Xenoliths

Minerals formed in the mantle and brought to the surface by eruptions. **Well-defined exposure time** between eruptions

- Available and **easy to retrieve** (LPC)
- Dating is easy thanks to eruptions
- Different samples from different volcanic areas

Risk: Low

If the prime samples are not available, alternative sources with **similar properties** but less interesting history can be identified and obtained

Costs for acquisitions are part of the 18k€ requested for retrieval missions.
Overheads can cover possible "paying" acquisitions.

Sample readout

The expected typical **size of the tracks** defines the best readout strategy

↓

Signal in the 1-100 µm range: can use the same techniques as fission tracks

Optical Microscopy

Accessible - Relatively cheap - Easily reproducible

Main requirements of the instrument:

Speed of scanning → Motorized stage and focusing

Ease and speed of **imaging** → Automated capture and transfer

Ease of adjustment → Modular design

Complete time availability



Modular microscopy station with motorized XY stage and autofocus from Thorlabs.
Cost of the proposal: 32 k€

Details of the proposal in the "PRIµS cost" document.

Risk: low

Delays or unavailability of components of the station can be replaced with other similar parts. In the case of bigger delays or malfunctioning, the chosen parts are versatile and adaptable to microscopes of many makes and models.

Sample preparation

The first preparation step for optical microscopy is **thinly slicing** the sample

Traditional workflow:

1. **Polishing** the sample with a diamond paste-based polisher
   → Expensive, extremely slow, wasteful
2. Enhancing the signal by enlarging the tracks via **chemical etching** (HF)
   → Needs dedicated safety protocols, non eco-friendly



Tabletop plasma cleaner by SPI supplies. Selected for stability and reproducibility of results.
Cost of the proposal: 11 k€

Details of the proposal in the "PRIμS cost" document.

Proposed alternative: the **Plasma Cleaner**

A streamlined **all-in-one solution**
→ Environmentally friendly!

Gas (Ar, O_2) is excited to a plasma state and ablates the surface of the sample cleaning from contaminants. Tracks are highlighted like in chemical etching.

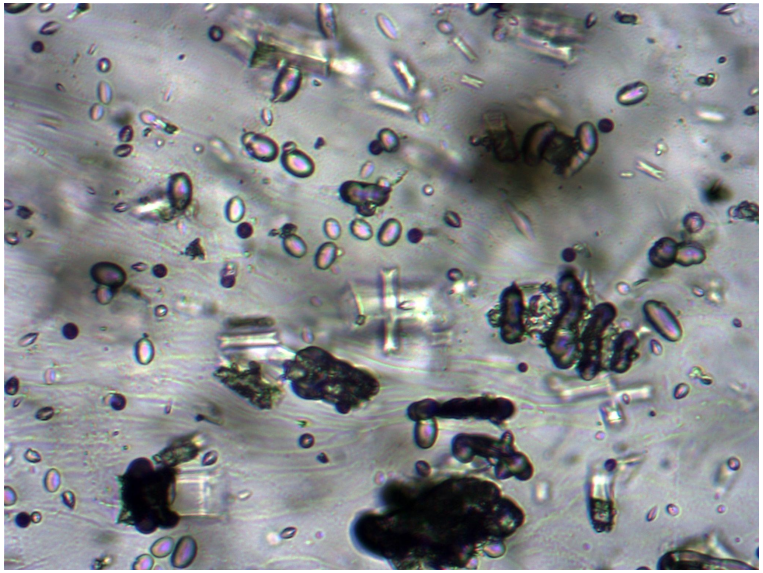
→ Already **demonstrated on halite.**

Risk: low
If the selected product is unavailable, there are other plasma cleaners at similar price points on the market. If there are malfunctions, a limited time share of similar equipment can be negotiated with other INFN groups.

Risk: low-medium
If the technique is not optimal for a set of selected samples, classical etching can be used.
Costs included in the overheads.

Image analysis

Once the images are acquired tracks have to be separated from other structural features, counted and their size must be estimated in a **pseudo-fourier analysis**.



Track tagging and measuring can be performed on different levels of automation

classical image analysis
to specialized AI tools

Ad-hoc trained **Mask-R CNN** or similar networks

Simple **OpenCV** ImageJ-based fourier Tools for cell counting

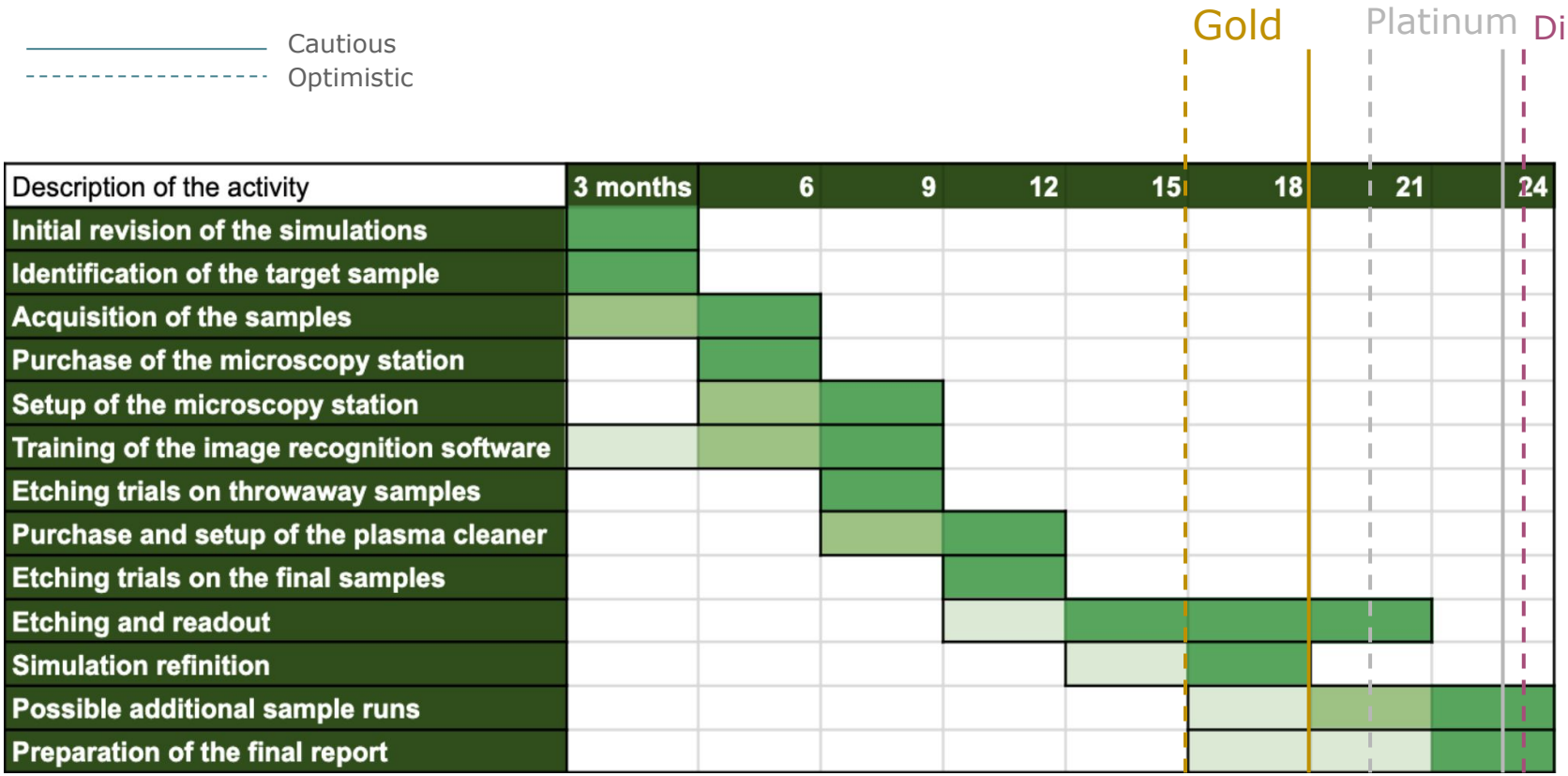
Overheads can cover paying for software

Calibration:
The fission peak can be used as the calibration signal.
Simulations will be refined after data acquisition.

Risk: low
If tracks in some samples are hard to tag with advanced tools, tagging can be done with simpler tools.

PRIµS project timeline

Cautious
Optimistic



The PRIμS group

INFN Milano and Università Statale di Milano		
Name	Position	FTE
Claudio Galelli (PI)	Post-doctoral Fellow	0.95
Lorenzo Caccianiga	Primo ricercatore	0.3
Lorenzo Apollonio	PhD Student (INFN associated)	0.4
-	Borsista INFN	0.7
Group at LPC - Université Clermont-Auvergne		
Vincent Breton	CNRS Directeur de recherche	0.15
Andreas Goudelis	CNRS Chercheur	0.1

The bulk of the technical work will be carried out in Milano.

The INFN Milano group:

- **Experience on sample preparation**, etching, microscopy, and track analysis with traditional techniques.
- **World-leading expertise** in cosmic ray track modeling within the paleodetector community.
- Multiple publications in the field.

The LPC group:

- **Strong connection to the geology-geophysics world**.
- Access to **samples**, including xenoliths and sediment from retrieval campaigns in the Mediterranean.

Budget recap and overview

Microscopy station:
Proposed model cost: 32k€
Requested funds: 33 k€

Primary working tool for scanning a large quantity of samples.
Main requirements: speed in scanning and imaging, modularity.

Plasma cleaner:
Proposed model cost: 11k€
Requested funds: 15 k€

More efficient than chemical etching, eliminates the use of harmful acids.
Main requirements: reproducibility, stability between different sample runs

Retrieval missions:
Requested funds: 18 k€

12 man-weeks total, for collaboration purposes (Clermont-Ferrand) and sample acquisition/direct retrieval or invites.

Overheads:
Requested funds: 5 k€

Unforeseen material purchases, consumables, external sample acquisition or preparations, software.

Total cost: 71 k€ + 10% contingency (inflation, price increases) = **78 k€**

PRIμS Summary

Objective: An **innovative detection technique** using optical microscopy to identify cosmic ray-induced ion tracks in ancient minerals.

Key Milestones:

- ***Gold*: Establish track-length spectrum standard for paleodetector research.**
- ***Platinum*: First confirmed identification of astroparticle-induced tracks.**
- ***Diamond*: Reconstruct historical cosmic ray flux variations from geological samples.**

Methods and Tools:

- Sample collection from ancient Halite deposits and/or volcanic xenoliths.
- High-resolution microscopy with automated scanning.
- Advanced etching and plasma cleaning for track enhancement.

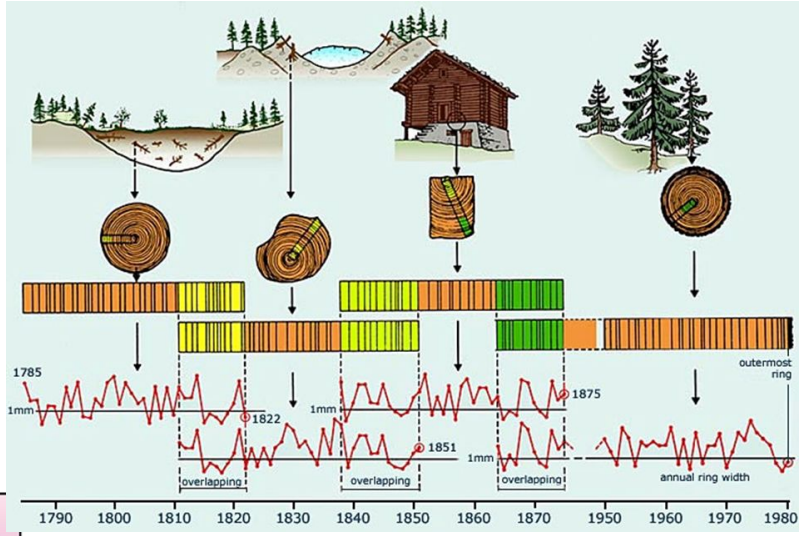
Impact: PRIμS aims to establish the **first experimental validation for paleodetectors**, bridging geology and high-energy astrophysics. It could transform the technique into a powerful tool - not just for cosmic rays, but also for dark matter and neutrino research - opening doors to **studying the deep history of cosmic events**. If the work is successful, we will be setting the stage for breakthroughs and birthing the field of **paleo-astroparticle physics.**

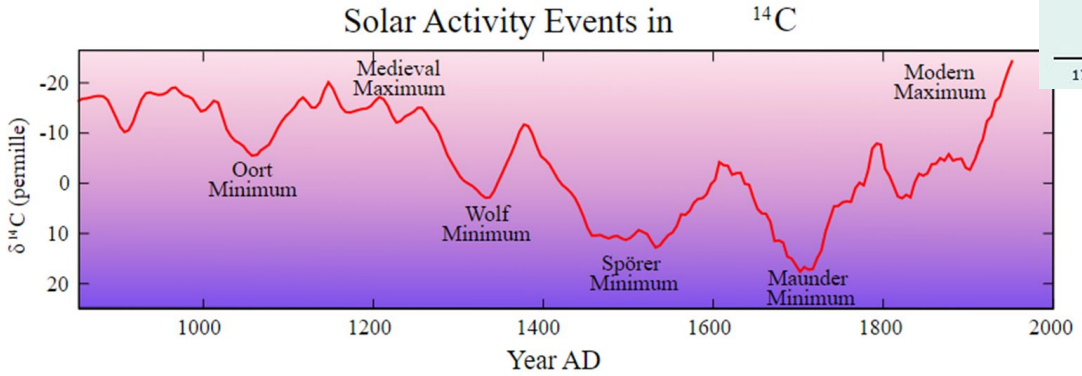
Thank you for your attention!



^{14}C evolution in atmosphere

- ^{14}C is produced by cosmic rays in atmosphere
- production is dominated by low energy cosmic rays that are strongly affected by solar activity
- the rate of ^{14}C is of huge interest for calibrating the dating technique



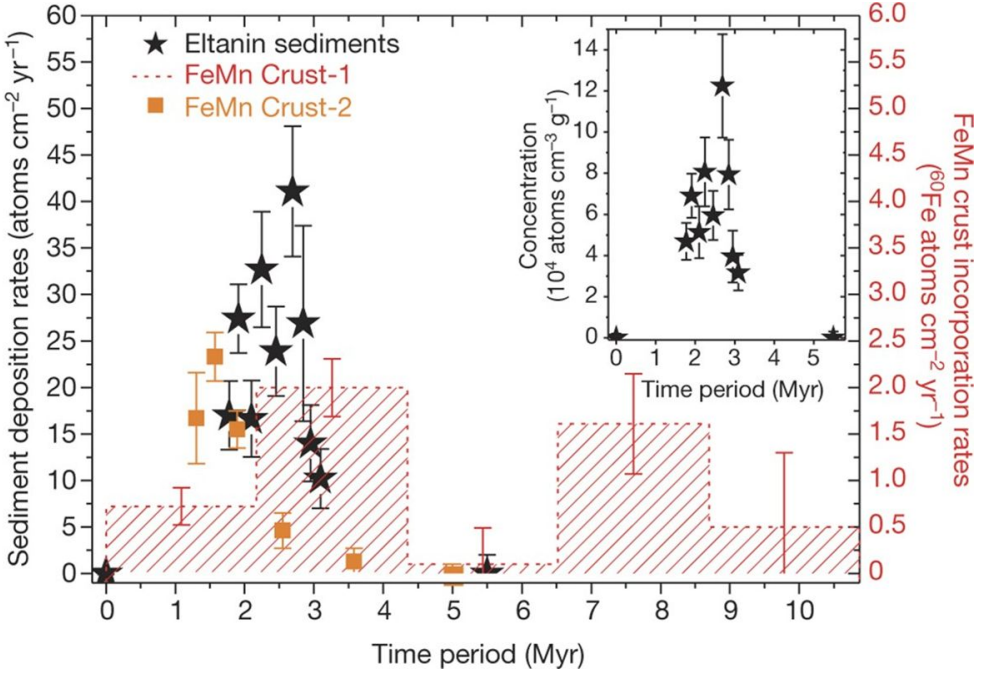


Traces of recent and close supernovae events

- ^{60}Fe deposit on oceanic crust
- Identified as trace of very close supernovae in the recent (few Myr) past

TABLE I: PREDICTIONS AND MEASUREMENTS OF ^{60}FE EXCESS IN DEEP OCEANIC CRUST SAMPLES (CORRECTED FOR IN SITU DECAY)

	Layer 1	Layer 2	Layer 3
age(Myr)	0-2.8	3.7-5.9	5.9-13
N_{SN}	8	4	6
D_{SN} (pc)	130	140	205
ϕ_{SN} (10^6cm^{-2} Myr^{-1})	$0.7^{+6.30}_{-0.06}$	$0.4^{+3.6}_{-0.04}$	$0.08^{+0.8}_{-0.01}$
ϕ_b(10^6 cm^{-2} Myr^{-1})	0.11	1.5	5
$\phi_{SN} + \phi_b$ (10^6cm^{-2} Myr^{-1})	$0.81^{+6.30}_{-0.06}$	$1.9^{+3.6}_{-0.04}$	$5.08^{+0.8}_{-0.01}$
ϕ_{obs} (10^6 cm^{-2} Myr^{-1})	$1.0^{+0.5}_{-0.3}$	8^{+11}_{-5}	$10^{+22}_{-8.5}$

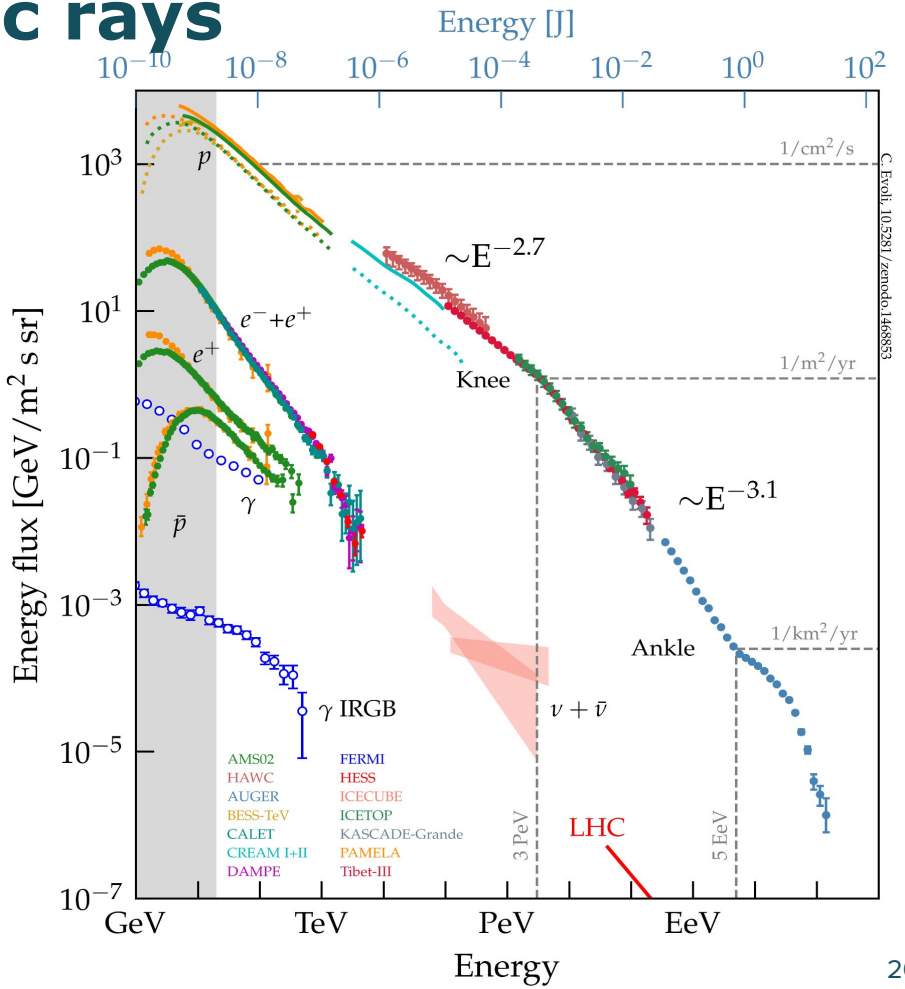


K. Knie, G. Korschinek, T. Faestermann, E. A. Dorfi, G. Rugel and A. Wallner, «60Fe Anomaly in a Deep-Sea Manganese Crust and Implications for a Nearby Supernova Source,» *Phyis. Rev. Lett.,* vol. 93, n. 17, p. 171103, 2004.

Cosmic rays

From few GeV to hundreds of EeV (1 EeV= 10^{18} eV)

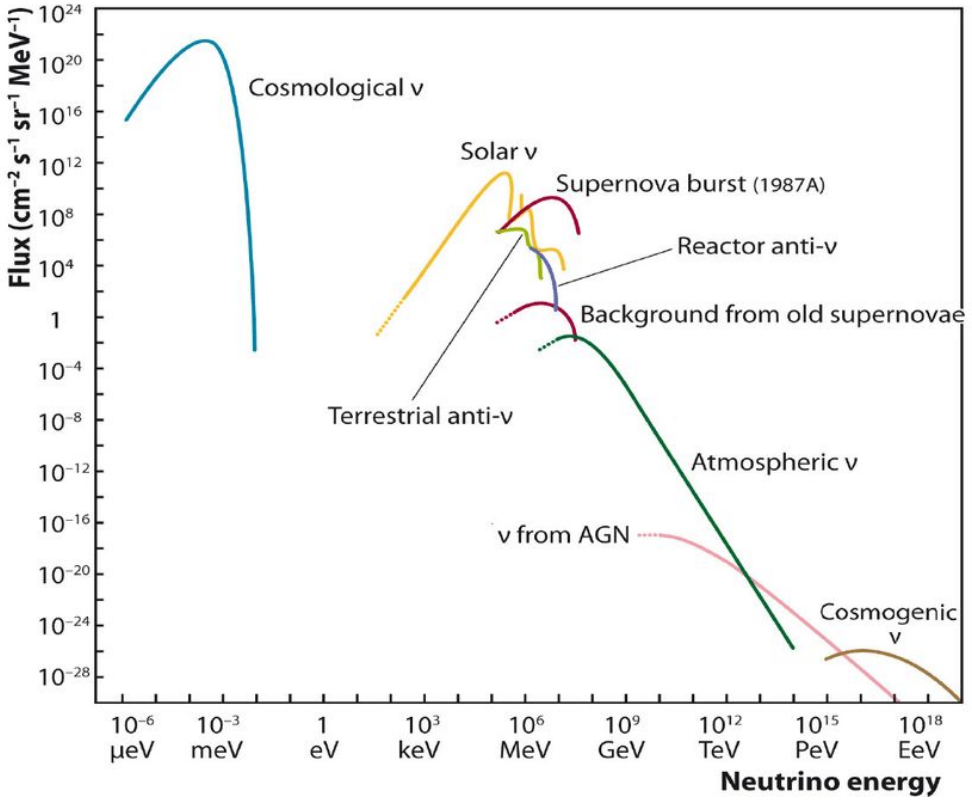
- **Most energetic particles known**
- Produce extensive air showers of **secondary particles** in the atmosphere: in these showers are muons, neutrons and neutrinos
- Primary cosmic rays mostly **charged particles** and are deflected by magnetic fields: delayed wrt to neutral particles
- cosmic rays in the TeV-PeV region associated to supernova remnants, higher energy may be accelerated by compact objects (magnetars?), AGN or starburst galaxies



Neutrinos

Difficult to reveal: their interaction is very unlikely

- need for large volumes
- at low energy main background is radioactivity
- Atmospheric neutrino background is practically ineliminable
- at the highest energies almost background-free but huge volumes (>km^3) needed

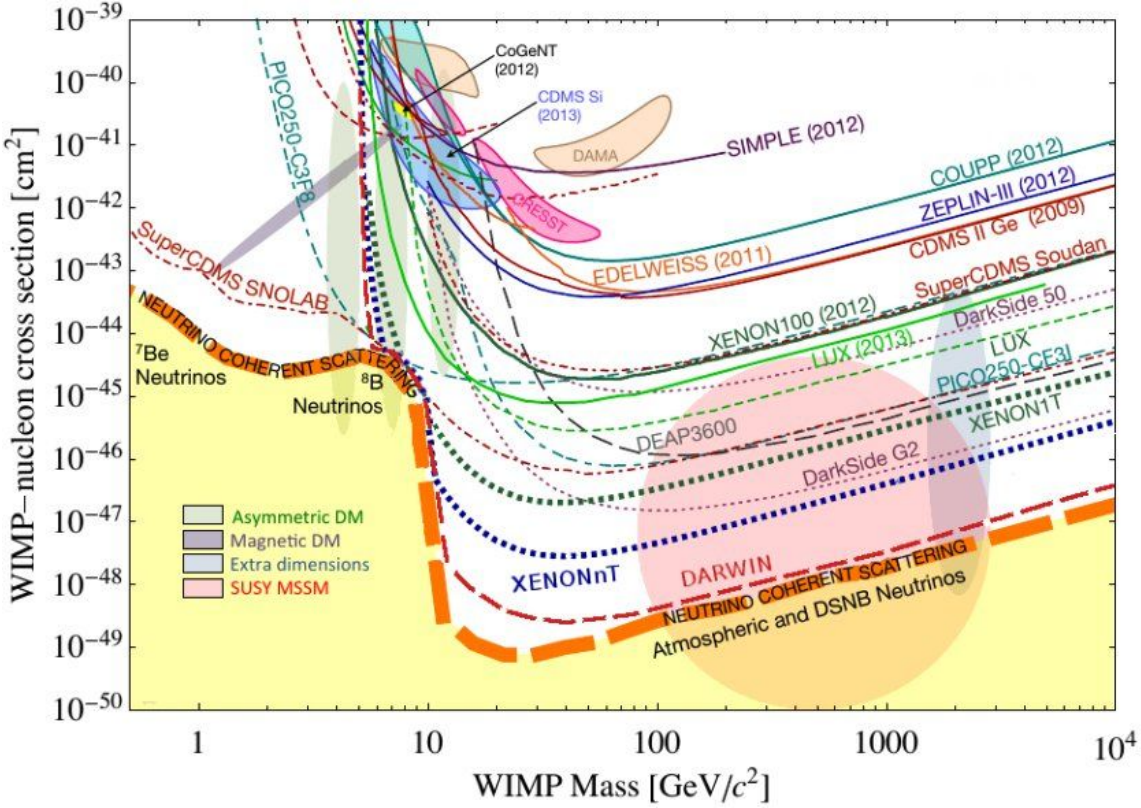


Dark matter

Direct detection based on the (supposed) flow of DM over earth. Main hypothesis is **WIMPs or axions**

Techniques rely on building **massive, ultrapure detecting environments** for rare events:

- noble scintillators (XENON)
- crystal scintillators (DAMA/LIBRA)
- cryogenic crystals (CRESST)



How do we detect rare events?

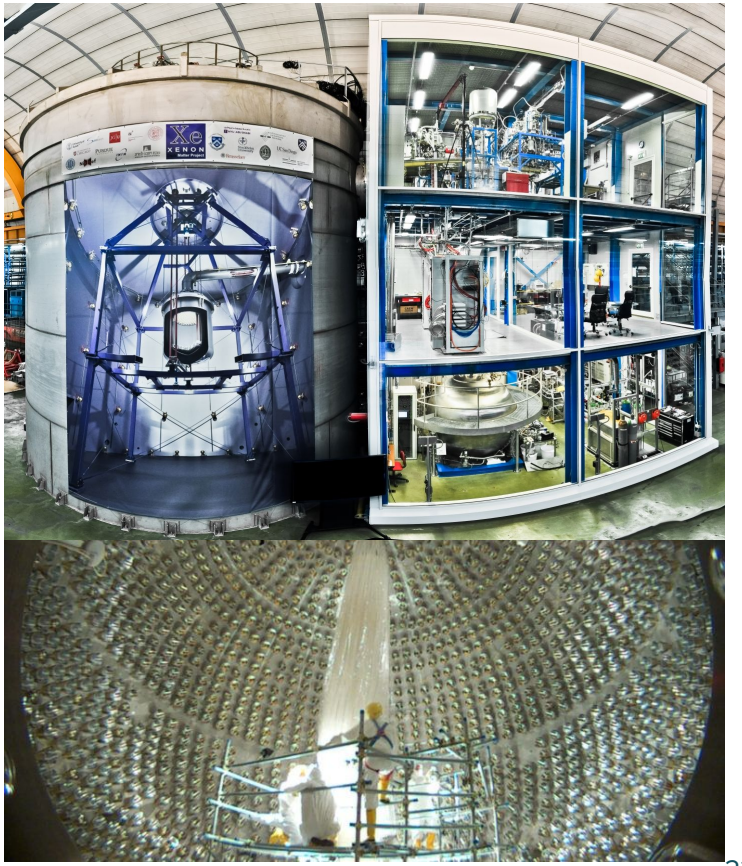
Is the event very rare and/or suppressed by background?

Create an environment in which **energy deposit from the desired interaction is distinguishable**

Internal background → Ultrapure material

External background → Efficient shielding

Rare event → Large detecting volume



Small detectors with a (very) long history

Minerals can be extracted from the Earth's crust with ages up to the order of Gyr

Xenon 1T
~1000 kg
~10 years data taking
Under ~ 1500 m of rock

Exposure= 10^4 kg yr





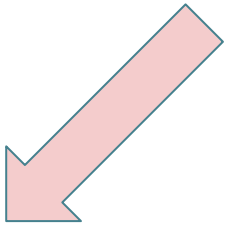
Morenosite crystals
~100 g
~1 Gyr
Under >2000 m of rock

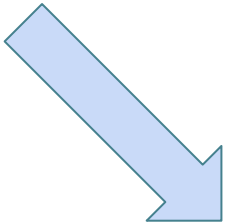
Exposure= 10^7 kg yr

Ok, but why?

Natural minerals can be (under certain conditions) particle detectors.

Why should we use them when we have much more refined detectors?





They can reach a very large exposure, even with a small sample

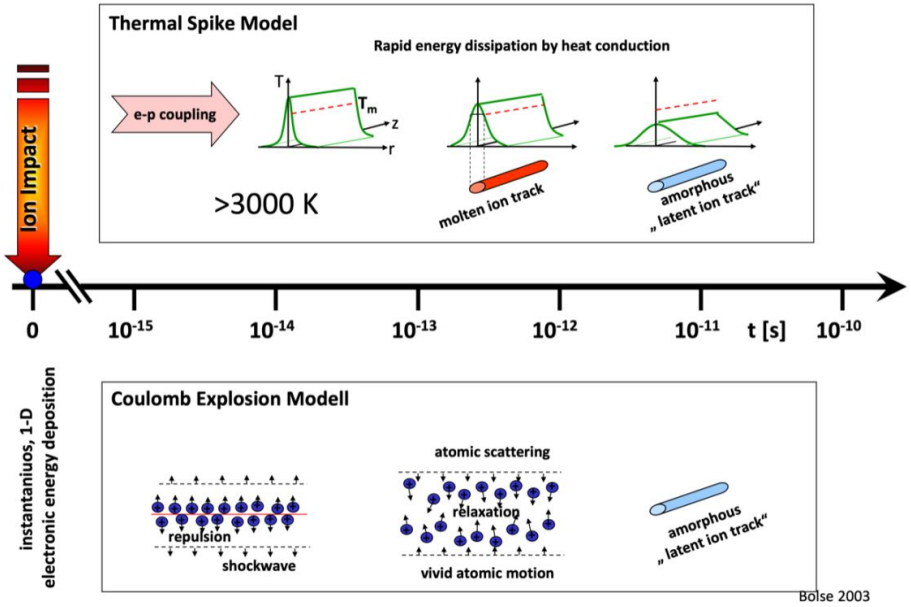
possibly larger than we could ever build artificially

They can measure the fluxes of astroparticles in the past

One of the very few chances of studying any astrophysical emission in the past

Choosing the Mineral
Generalities

The mechanism of track formation is not yet fully understood



A wishlist for the perfect paleo-detector candidate:

- **Insulators or poor semiconductors**
- **Low radioactive isotope content** to lower fission fragments contribution
- (Possibly) high H content to mitigate neutron background
- Easy to datate
- Easy to retrieve
- Abundant in different parts of the globe
- Forming often in Earth's history

Choosing the Mineral
Backgrounds: fission products and neutrons

Neutrons

From (α,N) reactions and spontaneous fission

Neutrons interact preferentially with hydrogen losing most of their energy
Hydrogen is too light to form tracks

More hydrogen = good

Fission fragments

Spontaneous fission from ^{238}U gives rise to two daughter nuclei that recoil back-to-back

Long - O(μm) double tracks

The more radiopure the better!

Minerals born in the **Mantle**

Evaporites from seawater

Choosing the Mineral
Geological history

The **history** of the mineral is as or more important than its composition

Dark matter and neutrino searches need very high overburdens to shield the minerals from secondary CRs (=muons)

Studies on the **evolution of the cosmic ray flux** need a source of samples with different ages but similar history

Searches for **transients** need samples of slightly different ages - born before and after the event

Studies on the **CR flux in a fixed moment** in the past need a sample that was exposed for a time and then shielded

These conditions often overwrite the preference on the mineral

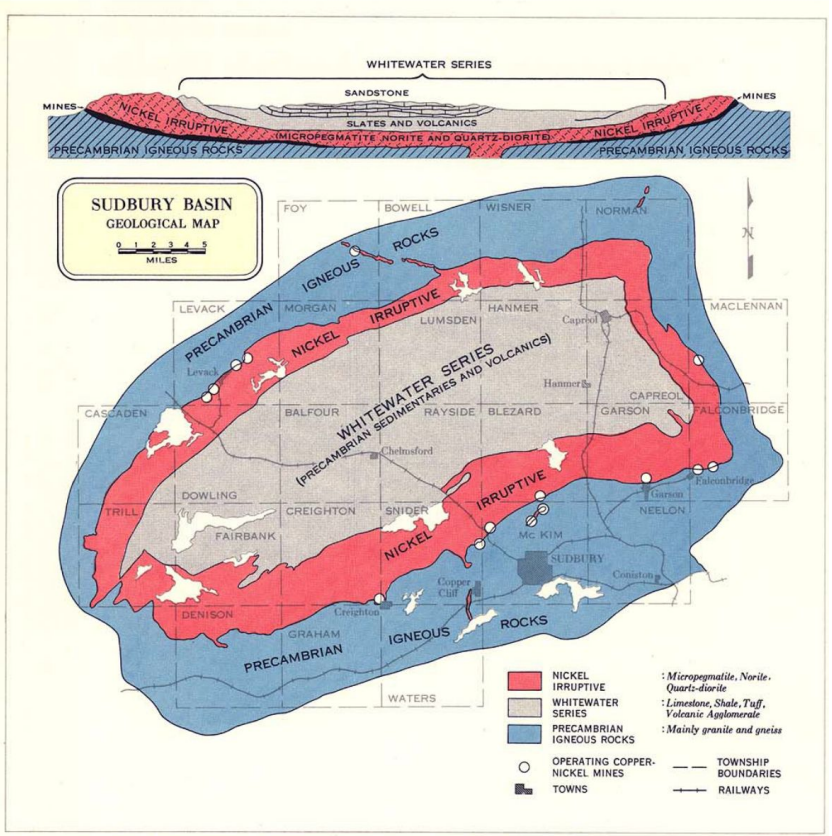
Geological history: examples of interest

The Sudbury region

Sudbury is a good candidate for paleo-detector extraction for searches that need high overburden, as it is recognised as the result of an asteroid impact that happened 1.85 Gyr ago: **well dated.**

It has many access points to **deep strata** in Nickel mines: excavation is possible

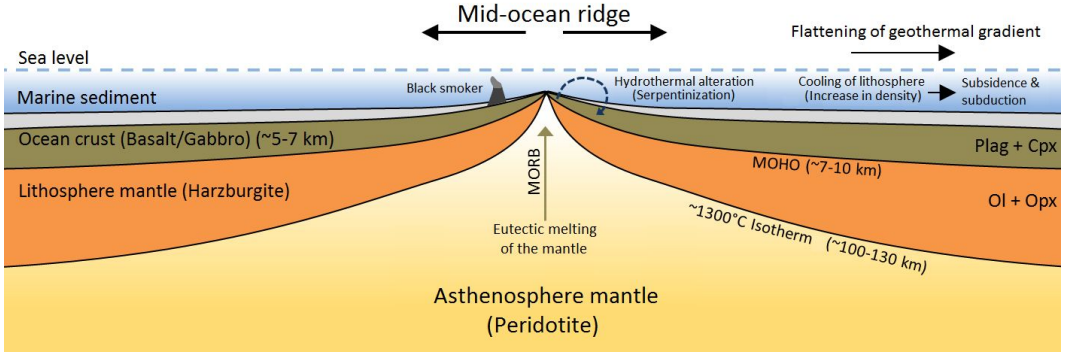
SNOLAB at 2km of depth is the deepest science facility in the world: it could provide on-site analysis possibilities, without the need of exposing the samples to present-day CR fluxes



Geological history: examples of interest

The Mid-Atlantic rift





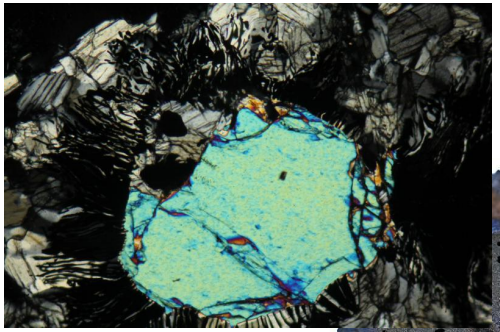
The continuous production of crust in the oceanic rift can be dated with paleomagnetism - could be exploited as a series of samples exposed for different integrated times.

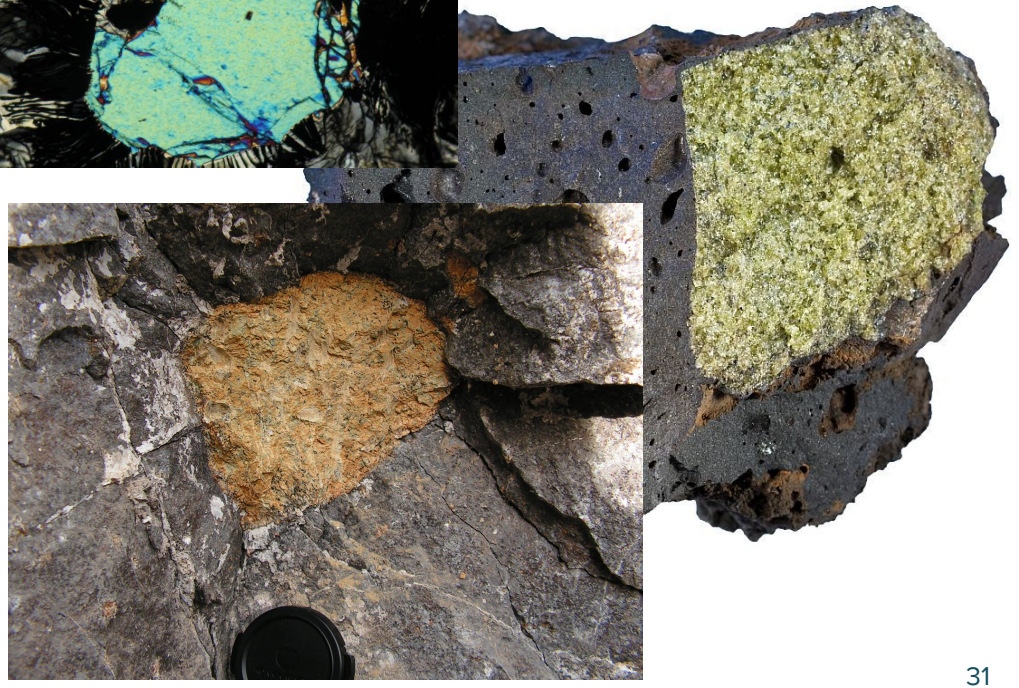
the Atlantic rift has different overburden of water, from ~4 km to practically none (iceland), which can be used to “select” high energy secondary particles, mostly produced by high energy primaries

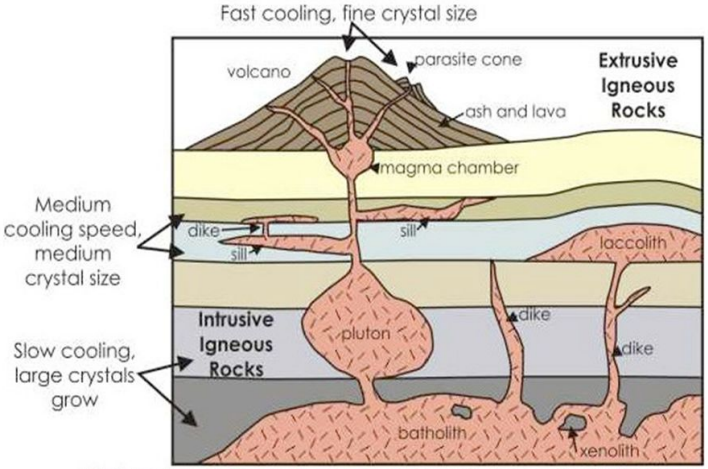
Geological history: examples of interest

Xenoliths and flux evolution

Xenoliths are intrusions into magmatic flows, taken mostly from the mantle and can easily be dated together with their associated eruptions.
Many types:
granodiorite, peridotite, sandstones, olivine...







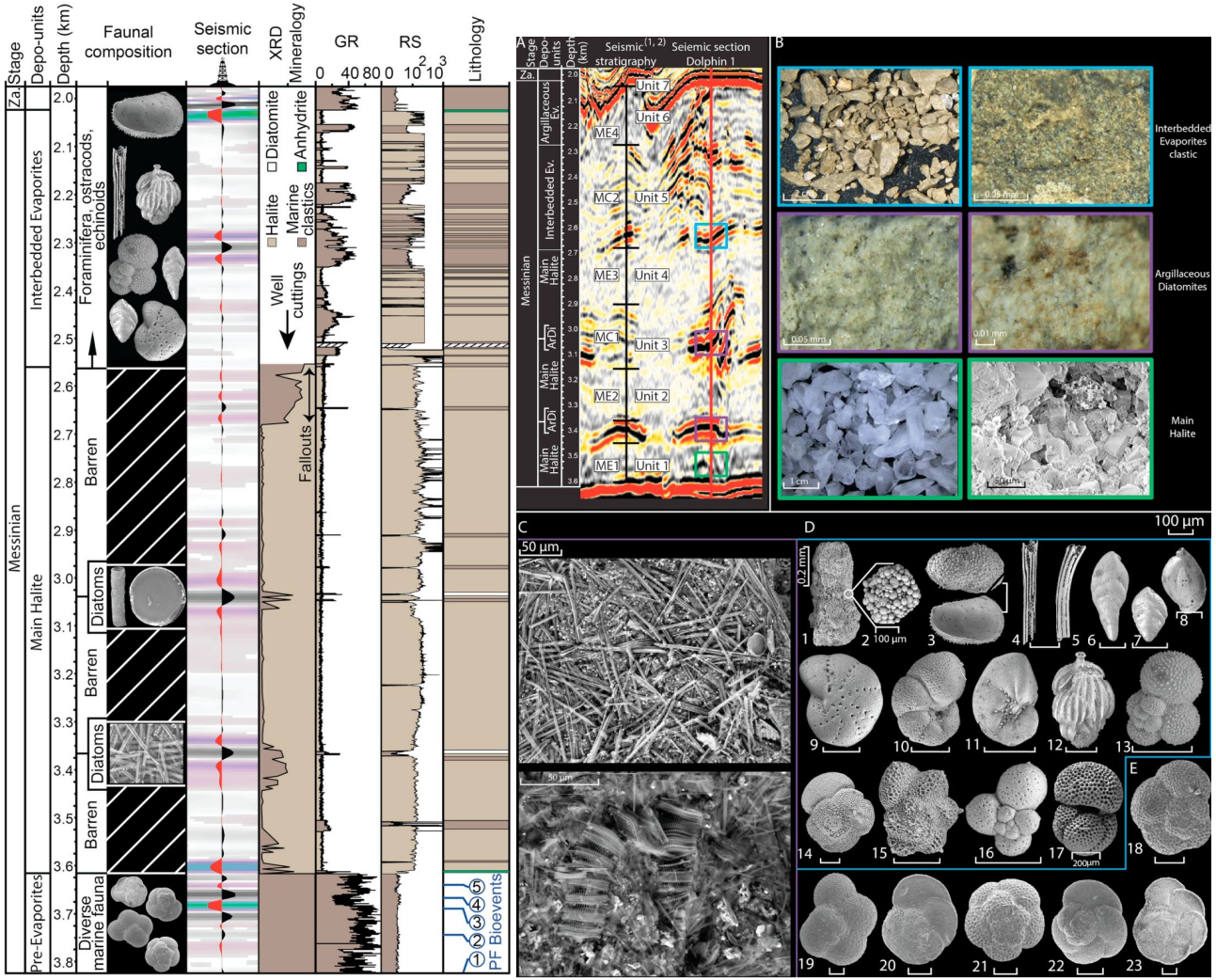
University of Canterbury, (unknown), taken 09/04/06 from http://outreach.canterbury.ac.nz/resources/geology/glossary/igneous.

Geological history: examples of interest
The Messinian Salinity Crisis and cosmic ray flux in the past

Main problem in paleo-detectors for cosmic rays is to know the **exposure** time of the mineral to the air:even a **small overburden** might change the flux dramatically.

Evaporites (Halite, Gypsum) produced in that event were exposed for a known period (~500 kyr) and then very briefly (possibly just **few years**) covered by a ~km of water





"Chronology with a pinch of salt: Integrated stratigraphy of Messinian evaporites in the deep Eastern Mediterranean reveals long-lasting halite deposition during Atlantic connectivity"
Meilijson et al.,
Earth-science reviews,
2019

The first proposals

Mica was one of the first targets (comes naturally in thin foils).

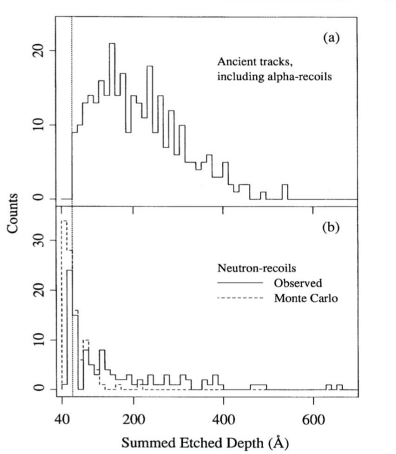
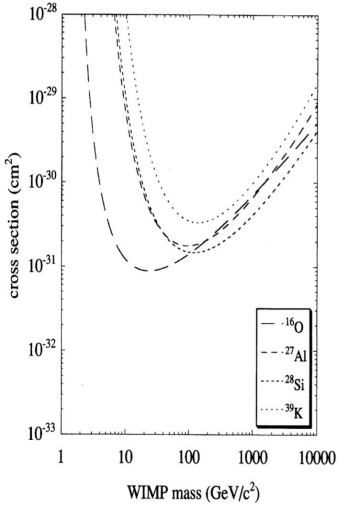


FIG. 3. (a) The summed etched depths of tracks recorded in a 80 720 μm^2 scan of 0.5 Gyr old muscovite mica. No events appear between our cutoff of 40 and 64 Å (shown with a dashed vertical line). (b) The solid line shows the summed depths of etched neutron-recoil tracks. The dashed line shows the results of a MC program of these data. In both the real and MC data a large fraction of the events appear in the 40–64 Å gap.



VOLUME 56, NUMBER 12 PHYSICAL REVIEW LETTERS 24 MARCH 1986

Search for Supermassive Magnetic Monopoles Using Mica Crystals

P. B. Price and M. H. Salamon
Department of Physics, University of California, Berkeley, California 94720
(Received 18 November 1985)

The observed absence of monopole tracks in large, ancient mica crystals enables us to set an upper limit of less than $\sim 10^{-18}$ cm^{-2} sr^{-1} s^{-1} on the flux of supermassive monopoles with $0.0004c < V < 0.0015c$ that are stably attached to nuclei. This limit takes into account the fraction of monopoles not initially bound to protons, the fraction that attach with nuclei on their way to the mica, and the measured storage times for tracks with thermal stability similar to that of monopole tracks.

PACS numbers: 14.80.Hv

VOLUME 74, NUMBER 21 PHYSICAL REVIEW LETTERS 22 MAY 1995

Limits on Dark Matter Using Ancient Mica

D. P. Snowden-Ifft,* E. S. Freeman, and P. B. Price*
Physics Department, University of California at Berkeley, Berkeley, California 94720
(Received 20 September 1994)

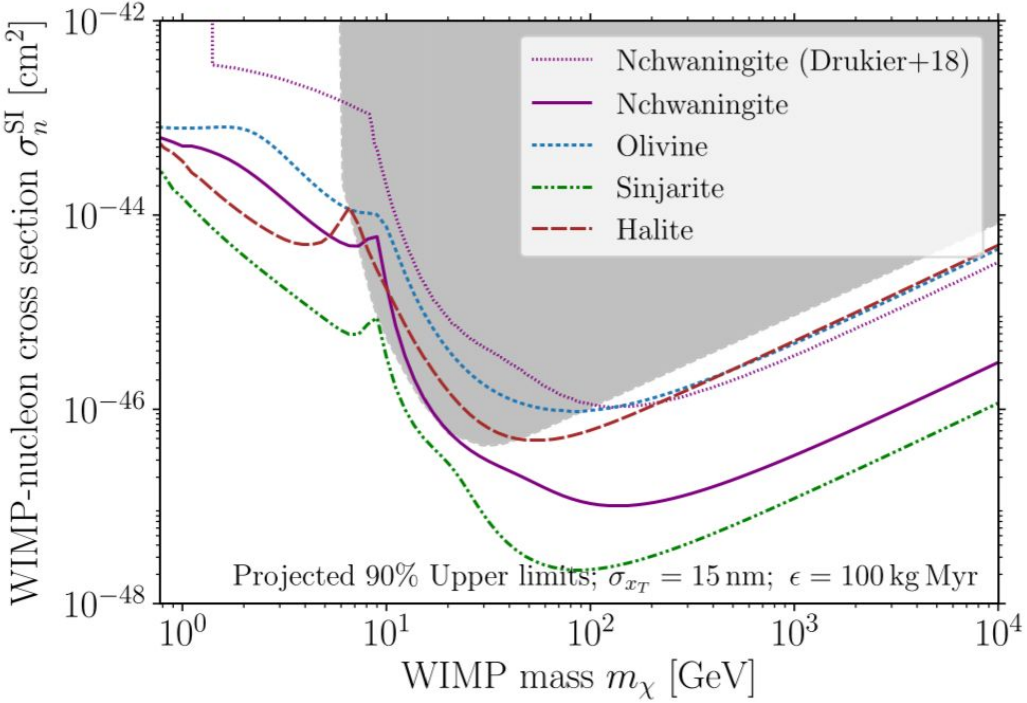
The combination of the track etching method and atomic force microscopy allows us to search for weakly interacting massive particles (WIMPs) in our Galaxy. A survey of 80 720 μm^2 of 0.5 Gyr old muscovite mica found no evidence of WIMP-recoil tracks. This enables us to set limits on WIMPs which are about an order of magnitude weaker than the best spin-dependent WIMP limits. Unlike other detectors, however, the mica method is, at present, not background limited. We argue that a background may not appear until we have pushed our current limits down by several orders of magnitude.

PACS numbers: 95.35.+d, 14.80.Ly, 29.40.Ym, 61.72.Ff

"Dark Horse" - Arons, 2003

Occasionally (once in $1/\nu_m^{(fast)} \sim 10^5$ years), such events must occur in our own galaxy, which would create a primary particle fluence at the earth of order 10^7 km^{-2}, for an average distance of 5 kpc. Such events might leave tracers in the geochemical record, a topic beyond the scope of this study.

2018 revival: paleo-detectors for dark matter detection

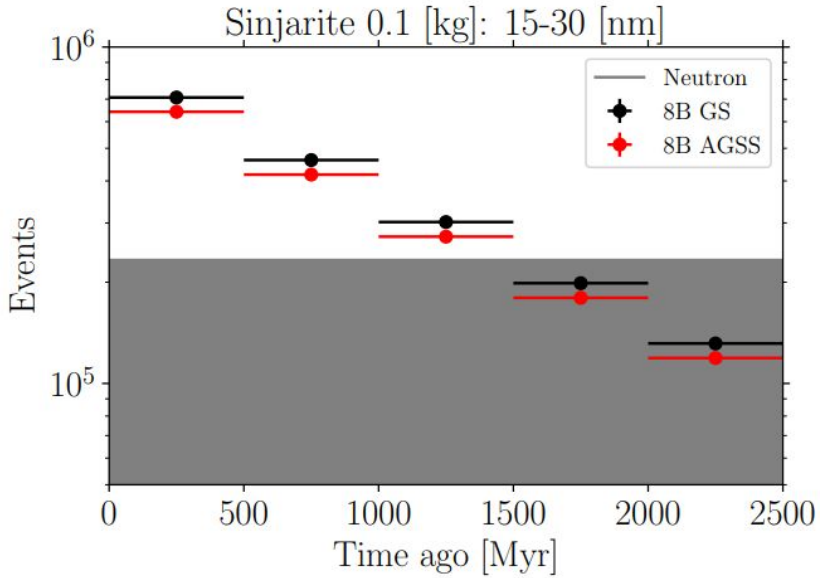


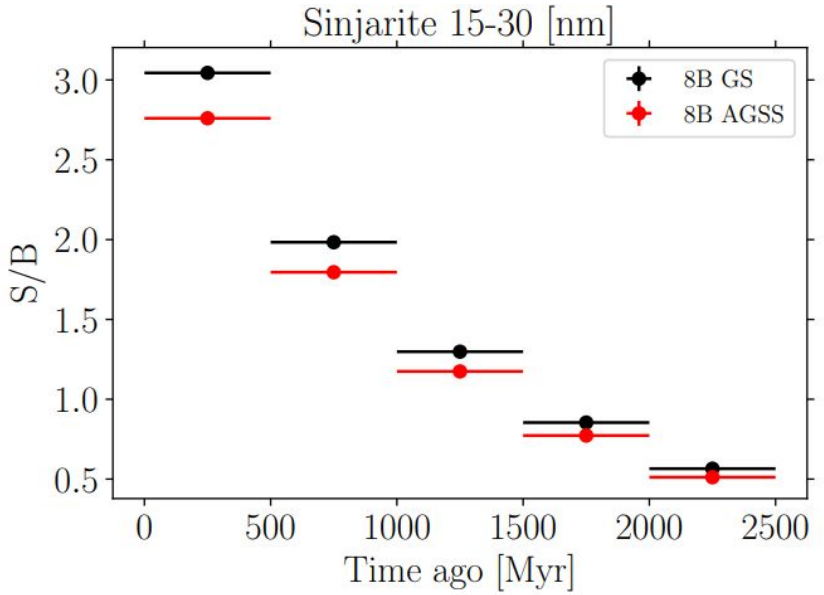
Projected paleo-detector 90% Upper Limits on WIMP-like DM. Assumes σ = 15 nm; = 100 kg Myr (high exposure case)

T. D. P. Edwards, B. J. Kavanagh, C. Weniger, S. Baum, A. K. Drukier, K. Freese, M. Gorski and P. Stengel, "Digging for dark matter: Spectral analysis and discovery potential of paleo-detectors," *Phys. Rev. D,* vol. 99, no. 4, p. 043541, 2019 arXiv:1811.10549

Paleo-detectors for solar neutrinos

- Paleo-detectors have been suggested also to study the flux of solar neutrinos to understand the sun metallicity
- Focus on ^{8}B neutrinos which are the most sensible to metallicity
- Collect samples at different ages might give you an idea of how the metallicity of the sun evolved

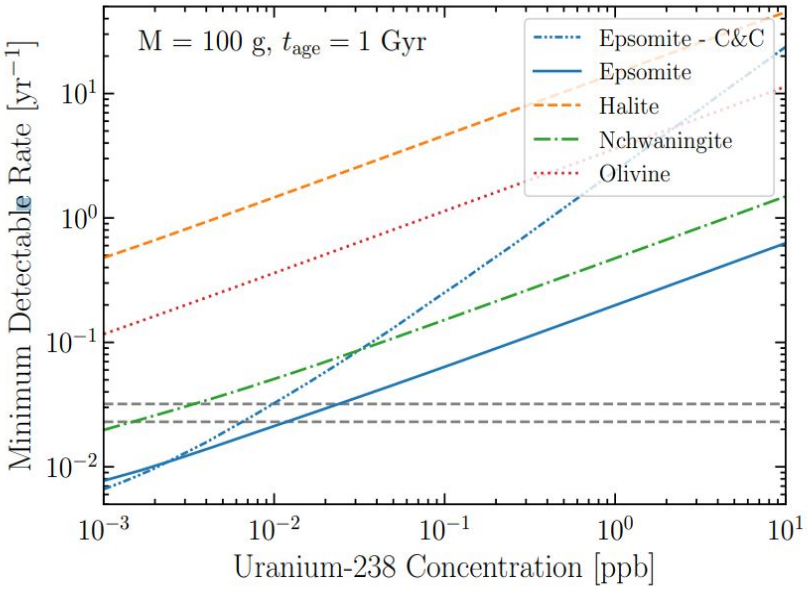


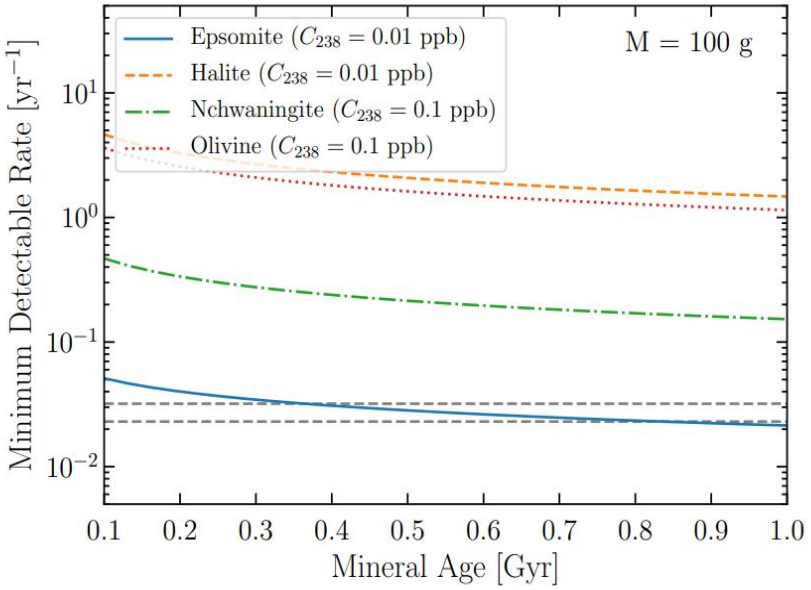


N. Tapia-Arellano and S. Horiuchi, "Measuring solar neutrinos over gigayear timescales with paleo detectors," *Physical Review D*, p. 123016, 2021

Paleo-detectors for supernova neutrinos

- Supernova neutrinos have a high instantaneous flux but for a very short (10s) duration
- Paleo-detectors might be able to measure the rate of SN in our Galaxy



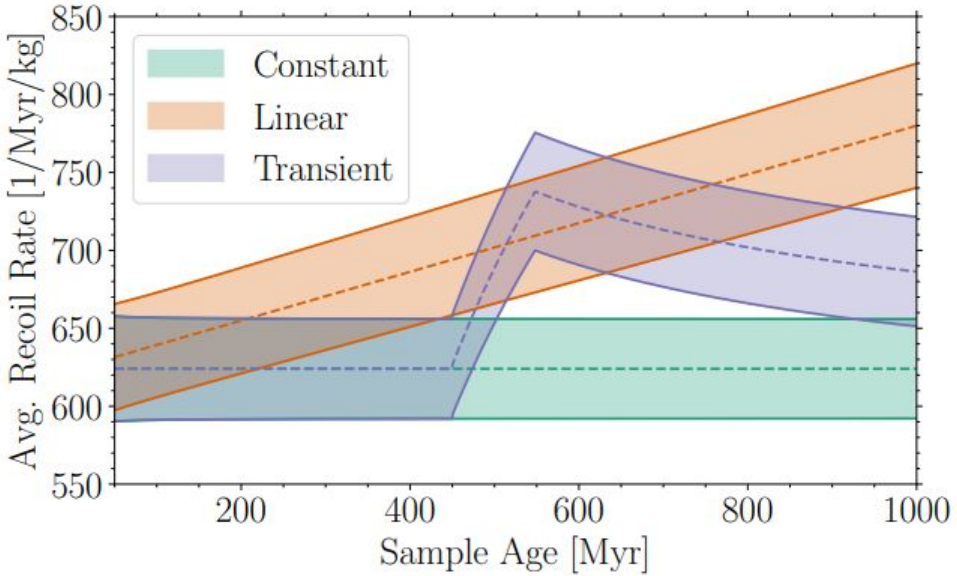


S. Baum, T. D. P. Edwards, B. J. Kavanagh, P. Stengel, A. K. Drukier, K. Freese, M. Gorski and C. Weniger, «Paleodetectors for Galactic supernova neutrinos,» *Phys. Rev. D*, vol. 101, n. 10, p. 19, 2020

Paleo-detectors for cosmic rays

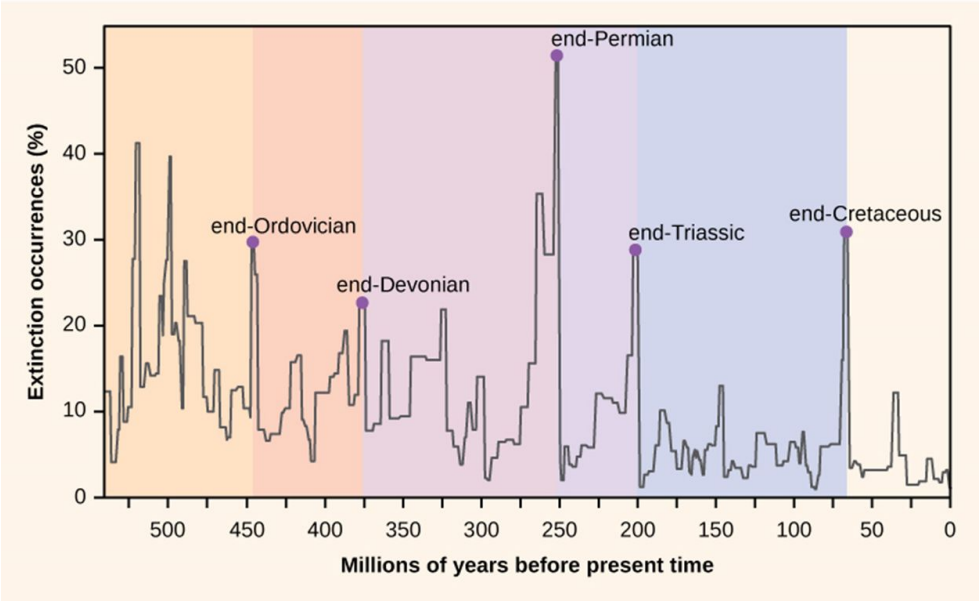
- Atmospheric neutrinos might be able to trace an evolution of the flux of cosmic rays and search for long transients (100 Myr)
- For shorter transients a different approach might be better (see after)
- The ability of measuring **CR transients** depends on where the transients are located: cosmic rays are charged particles and are thus **delayed** by magnetic fields
- Depending on their **rigidity**, they might even be trapped inside the Galactic Magnetic Field (GMF)
- Exception: if sufficiently **close and/or energetic** neutrons can arrive to Earth.
  *decay length ~9 kpc * (E [EeV])*

J. R. Jordan, S. Baum, P. Stengel, A. Ferrari, M. C. Morone, P. Sala and J. Spitz, «Measuring Changes in the Atmospheric Neutrino Rate over Gigayear Timescales,» *Phys. Rev. Lett.,* vol. 125, n. 23, p. 231802, 2020

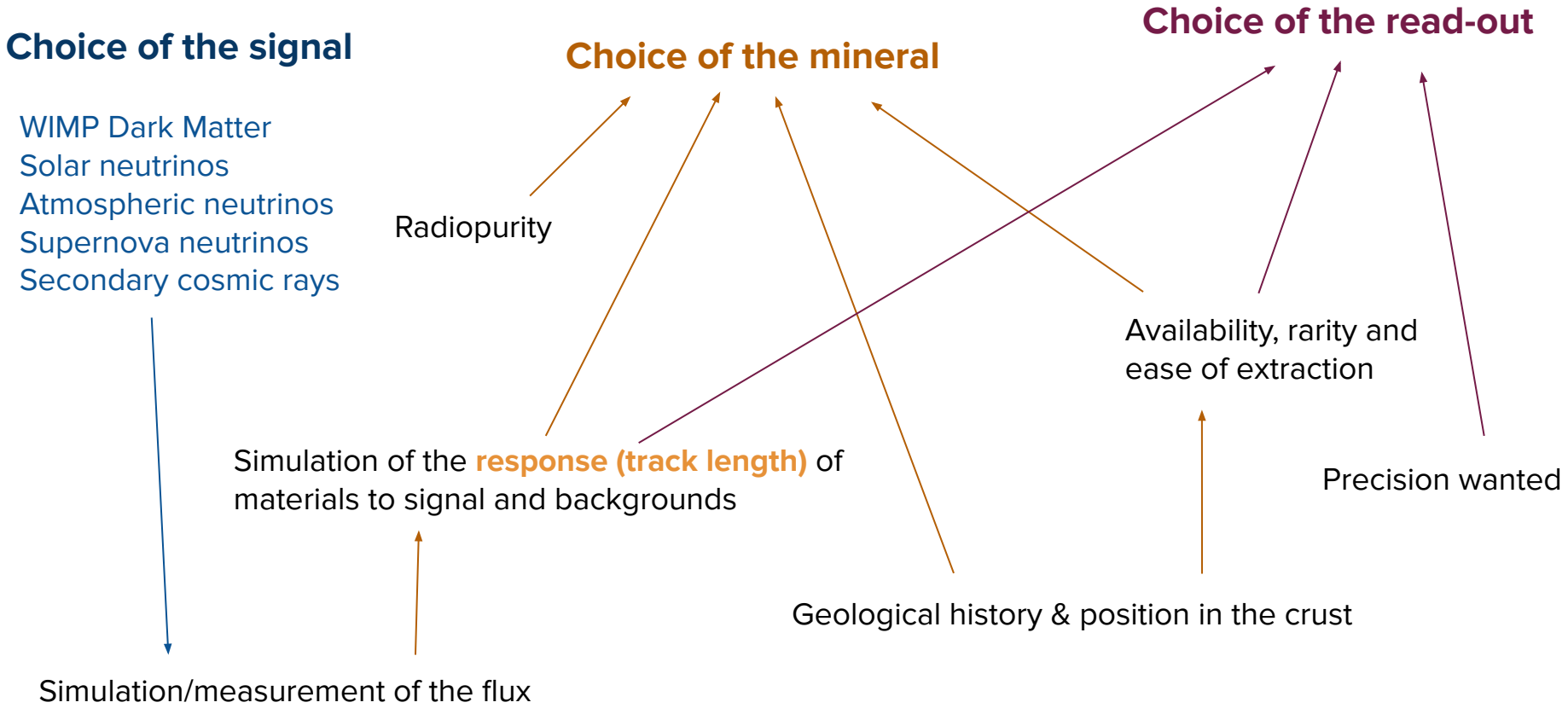


Mass extinctions: caused by nearby astrophysical events?

- It was suggested in various occasion that a **nearby catastrophic astrophysical event** might be the cause of some of the various **mass extinctions** that happened on Earth
- **gamma** emission from SN or GRB could have **depleted the ozone layer**, causing harmful UV radiation to kill surface-living species (or making them evolve quicker)
- some extinctions show however that also **deep-sea** and **deep-underground** creatures were affected. It was suggested that it might be the effect of elevated **muon** (and/or **neutron**?) fluxes induced by a burst of primaries
- such a flux might easily have left **tracks in paleo-detectors**



The recipe for paleo-detectors



Simulations: the paleopy package

https://github.com/tedwards2412/paleopy

Thomas Edwards
Bradley Kavanaugh

paleopy computes the **track length spectrum** for different sources and backgrounds in selected materials

```
class Mineral:
    def __init__(self, mineral):
```

It defines the characteristics of the mineral: number of different nuclei, mass/charge of the components, molar mass
Currently supported: nchwaningite, halite, epsomite, nickelbischofite, olivine, sinjarite + morenosite, obisidian (in progress)

```
def loadNeutronBkg(self):
```

Computes backgrounds for the selected mineral: neutrons, fission products

→ From SOURCES4A, SRIM

```
def dRdx_nu(self,x_bins, components=False, gaussian=False):
nu_list = ['DSNB', 'atm', 'hep', '8B', '15O', '17F', '13N', 'pep','pp','7Be-384','7Be-861']
```

Computes track lengths for neutrinos of different sources:
can be used for other particles

→ From Corsika, Geant & other software

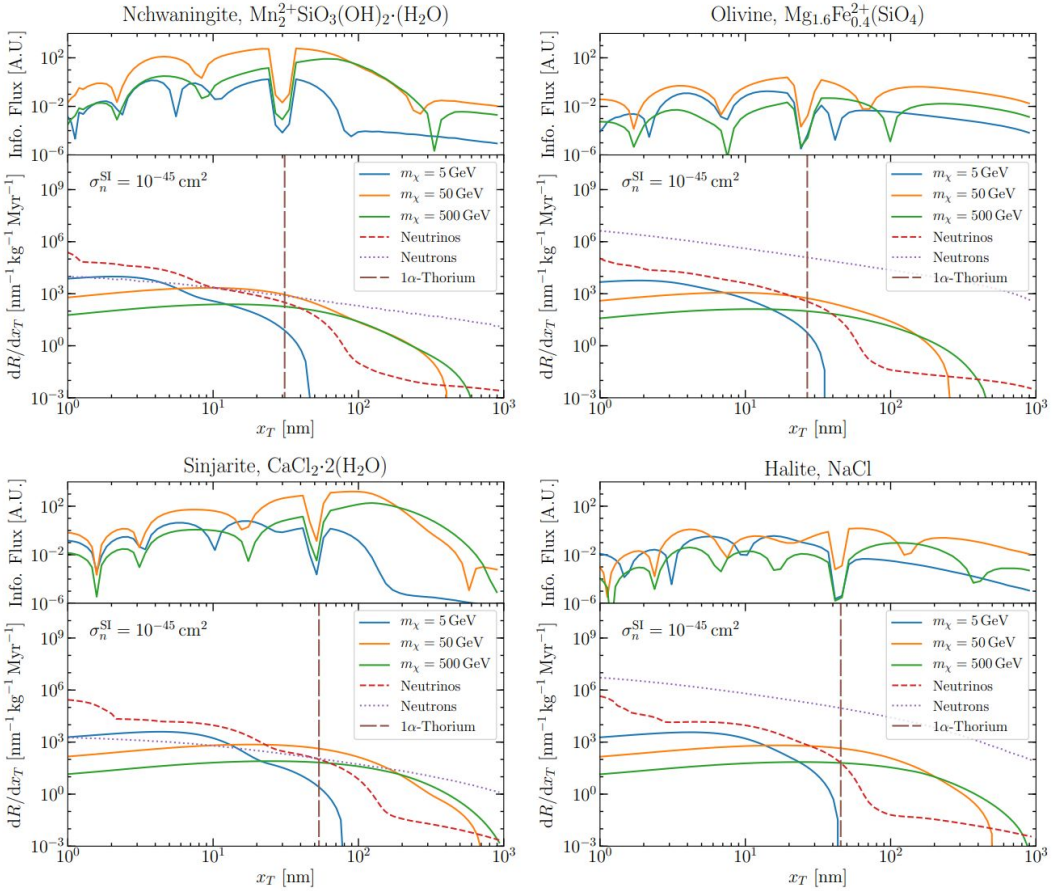
```
def dRdx(self, x_bins, sigma, m, gaussian=False):
```

Computes track length for DM of mass m

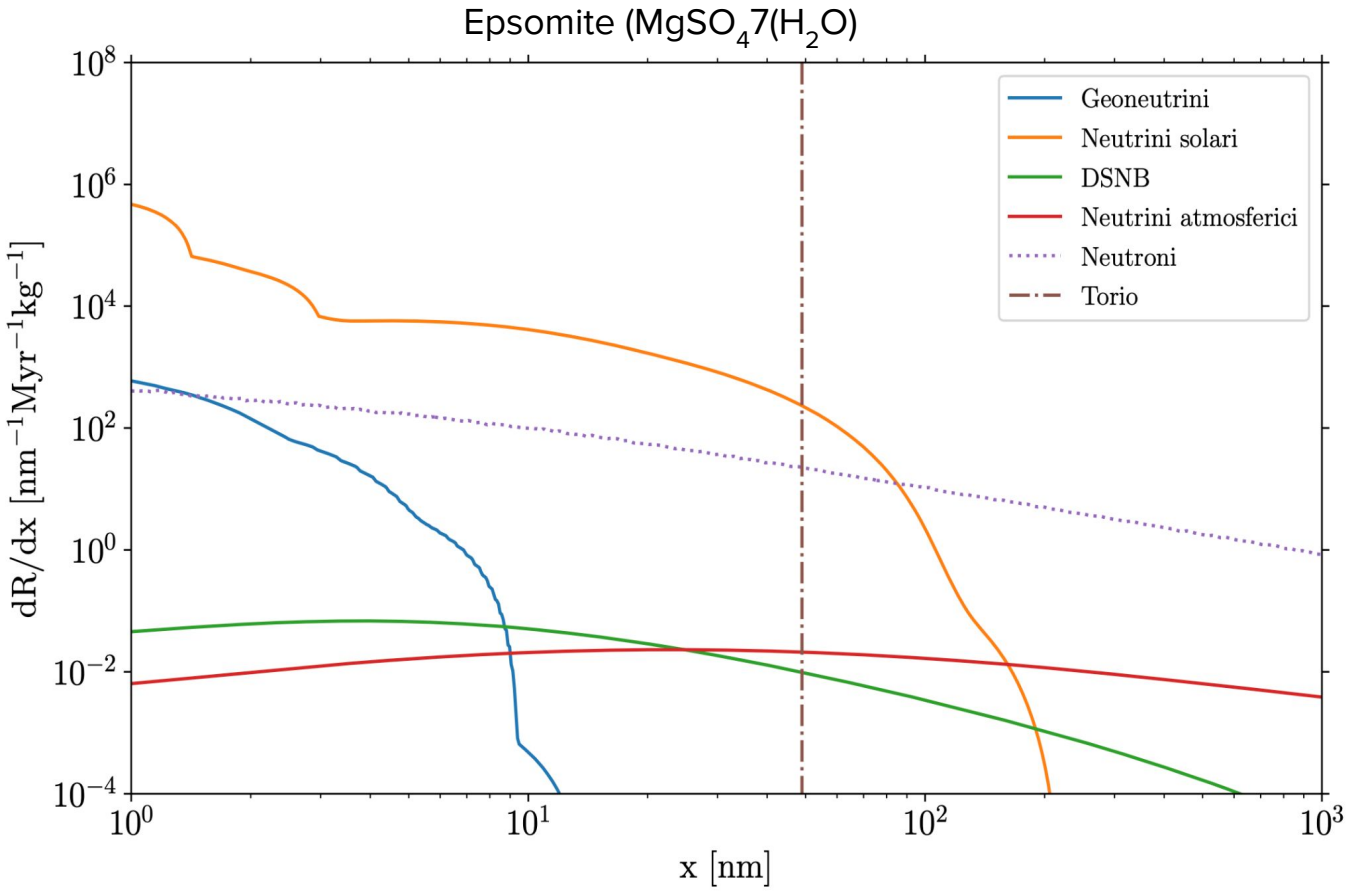
→ From WIMPy_NREFT

Simulations: the paleopy package

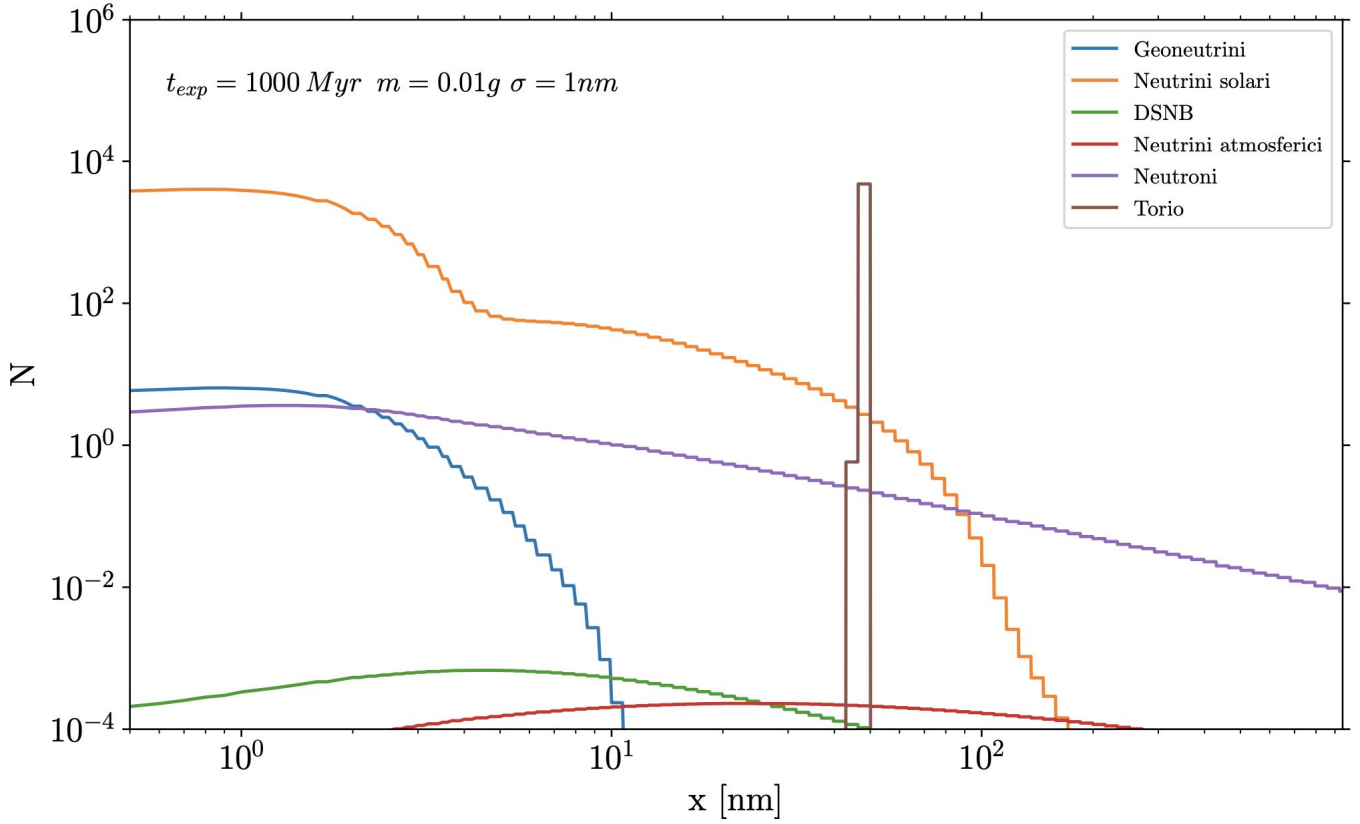
T. D. P. Edwards, B. J. Kavanagh, C. Weniger, S. Baum, A. K. Drukier, K. Freese, M. Gorski and P. Stengel, "Digging for dark matter: Spectral analysis and discovery potential of paleo-detectors," *Phys. Rev. D,* vol. 99, no. 4, p. 043541, 2019



Simulations: the paleopy package



Simulations: the paleopy package

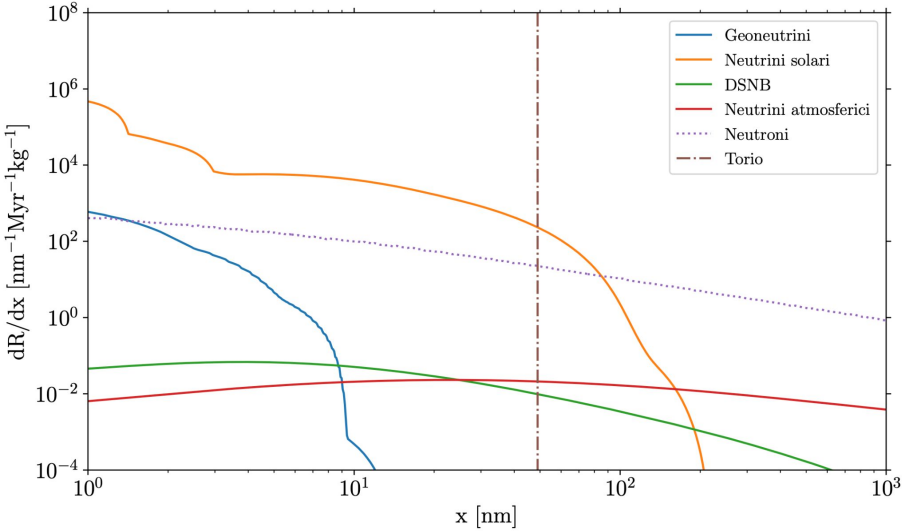


Case study 1: neutrinos from galactic supernovae

Name / description	Age	Distance (ly)	Time suppression wrt steady sources (10s emission)	Distance enhancing factor wrt SN1987A	Total suppression coefficient wrt steady sources
SN1987A	34 yr	168,000	9.3×10^{-9}	1	9.3×10^{-9}
Vela jr	800 yr	700	4.0×10^{-10}	5.8×10^{4}	2.3×10^{-5}
Geminga	342 kyr	815	9.3×10^{-13}	4.3×10^{4}	4.0×10^{-8}
Vela	11 kyr	815	2.8×10^{-11}	4.3×10^{4}	1.2×10^{-6}
Crab (SN1054)	967 yr	6300	3.3×10^{-10}	7.1×10^{2}	2.3×10^{-7}
SN1572	449 yr	7500	7.1×10^{-10}	5.0×10^{2}	3.5×10^{-7}
SN 1006	1015 yr	7200	3.1×10^{-10}	5.4×10^{2}	1.7×10^{-7}
Possible very close SN from ^{60}Fe deposits [11]	2.8 Myr	130.4	1.1×10^{-13}	1.7×10^{6}	1.9×10^{-7}
20 explosions in 40-130 pc in the last 11 Myr [27]	11 Myr	327	2.9×10^{-14}	2.6×10^{5}	1.5×10^{-7}
8 SN at around 130 pc in the last 2.8 Myr [27]	2.8 Myr	425.1	1.1×10^{-13}	1.6×10^{5}	3.5×10^{-7}

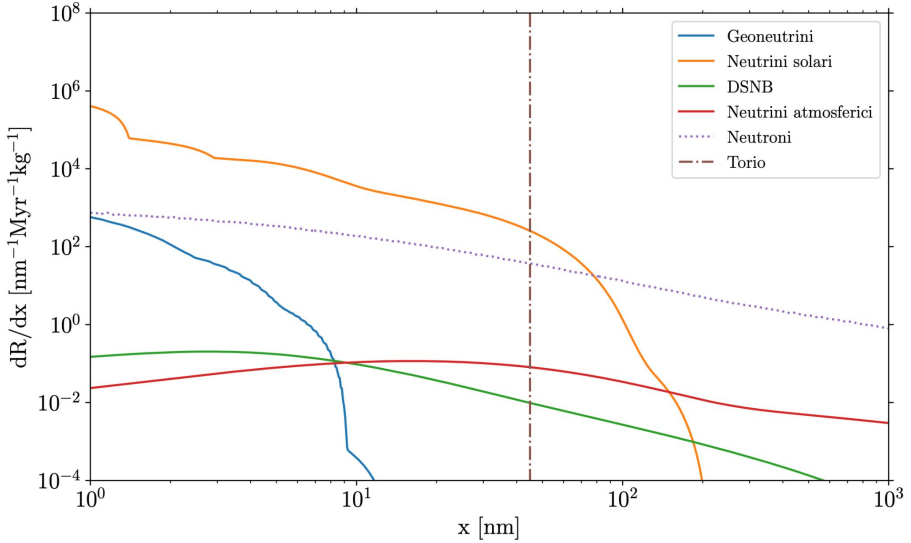
Case study 1: neutrinos from galactic supernovae

Epsomite ($MgSO_4 7(H_2O)$)



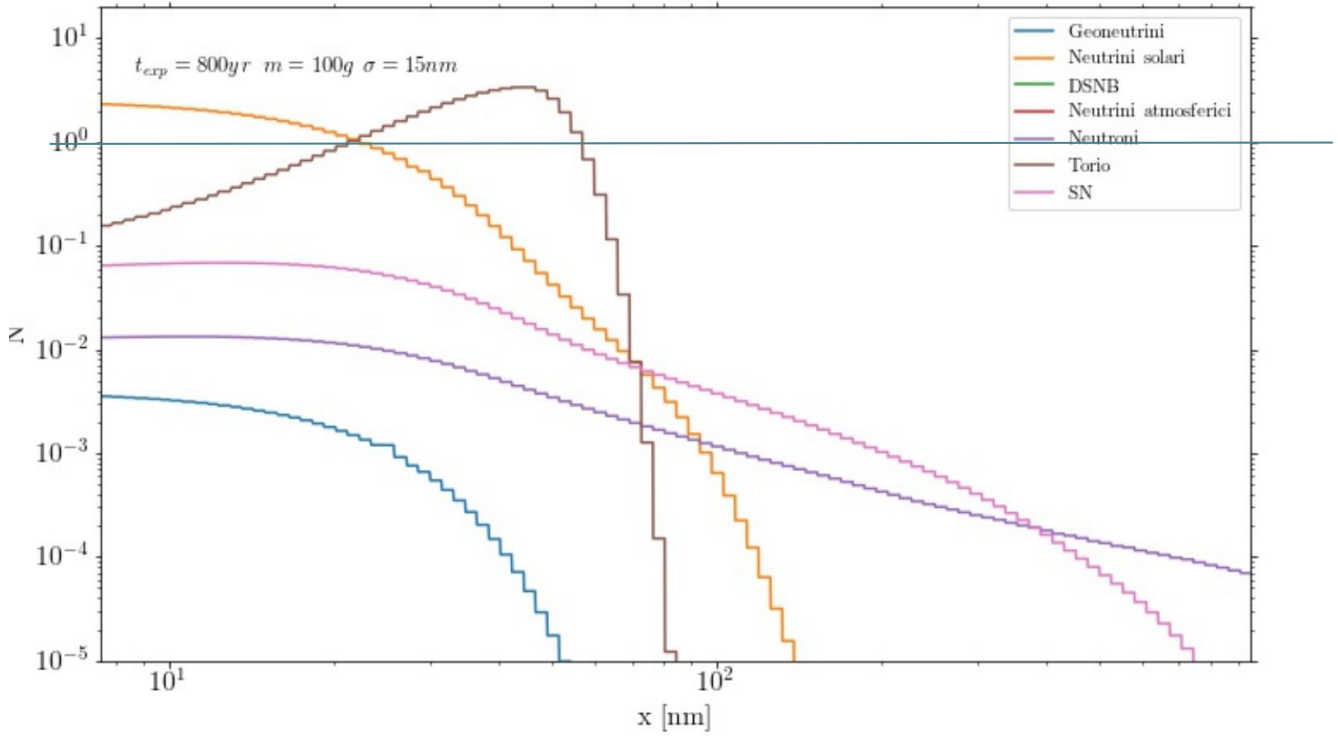
Hard to obtain with high enough overburdens: mostly found in carsic sediments and dried salt basins

Morenosite ($NiSO_4 7(H_2O)$)



Found in Sudbury's Nickel mines

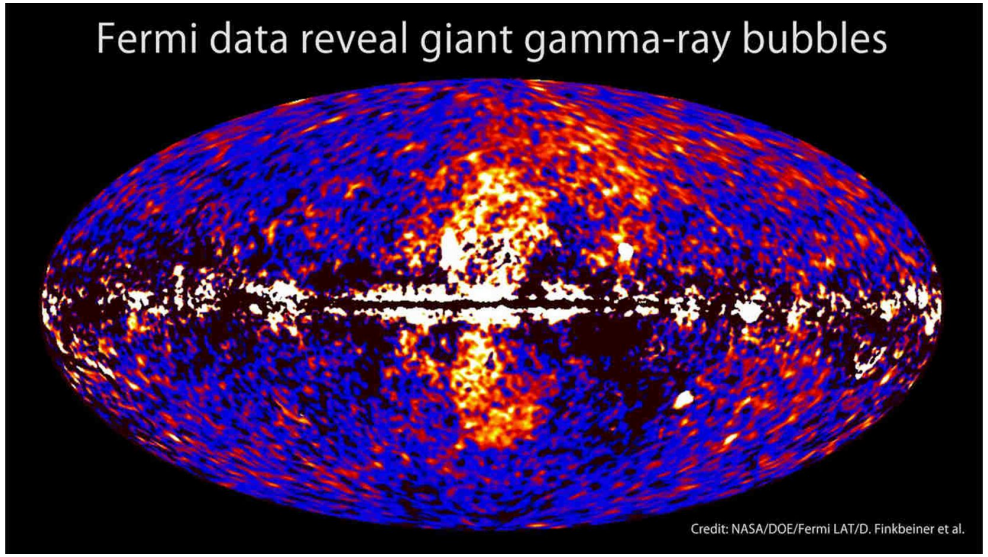
Case study 1: neutrinos from galactic



Huge volume needed for detection! - even in the best case scenario

A possible case study: Fermi Bubbles (and their age)

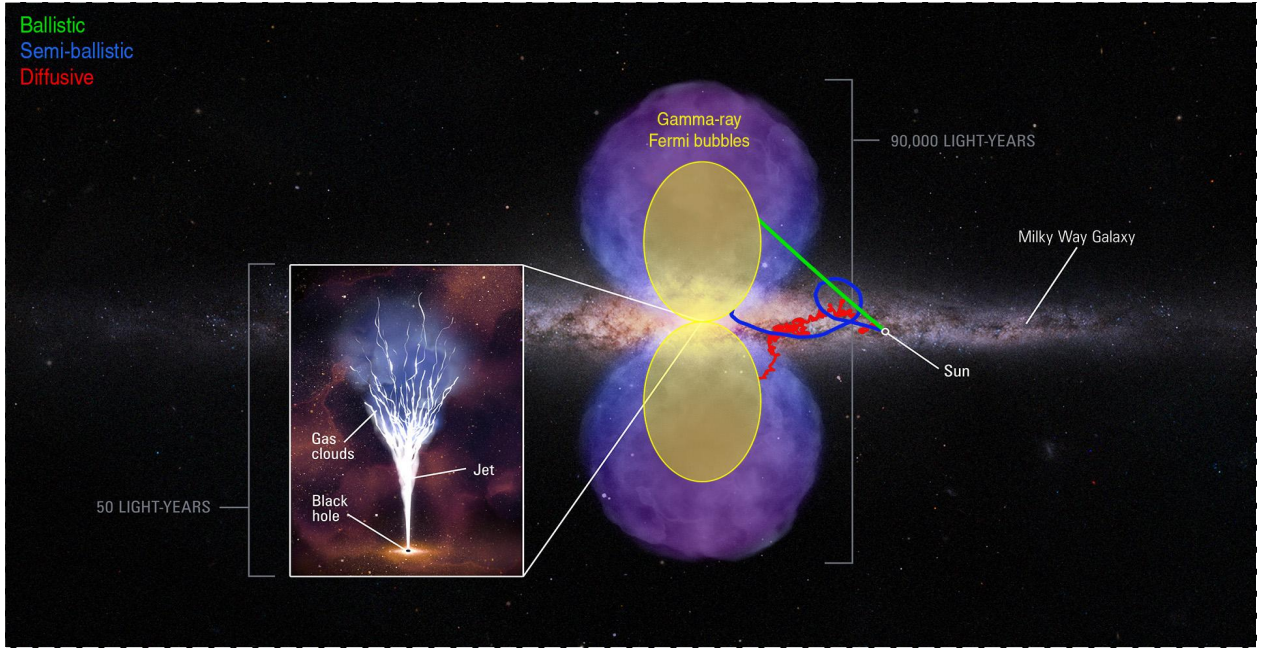
- The observation of **bubble-like** structure perpendicular to the galactic plane in gamma showed that our Galaxy might have been active in the recent past
- Is it possible that this activity was associated to **high energy cosmic ray emission**?
- Some studies date these structures at **5-6 Myr ->** nicely fits the period where the Mediterranean was **mostly evaporated (5.9 Myr)**



M. Su and D. P. Finkbeiner, "EVIDENCE FOR GAMMA-RAY JETS IN THE MILKY WAY," Astrophys. Journal, vol. 753, no. 61, 2012.
R. Zhang and F. Guo, «Simulating the Fermi Bubbles as Forward Shocks Driven by AGN Jets,» Astrophys. Journal, vol. 894, n. 117, 2020.
M. Su, T. Slatyer and D. Finkbeiner, «GIANT GAMMA-RAY BUBBLES FROM FERMI-LAT: ACTIVE GALACTIC NUCLEUS ACTIVITY OR BIPOLAR GALACTIC WIND?,» Astrophys. Journal., vol. 724, p. 1044, 2010.

Case study 2: MSC and the Fermi bubbles

Phase 1: Simulate UHECR emission from the Fermi Bubbles, propagate and check which emitted particles can affect evaporites from the MSC before they are shielded by the flood

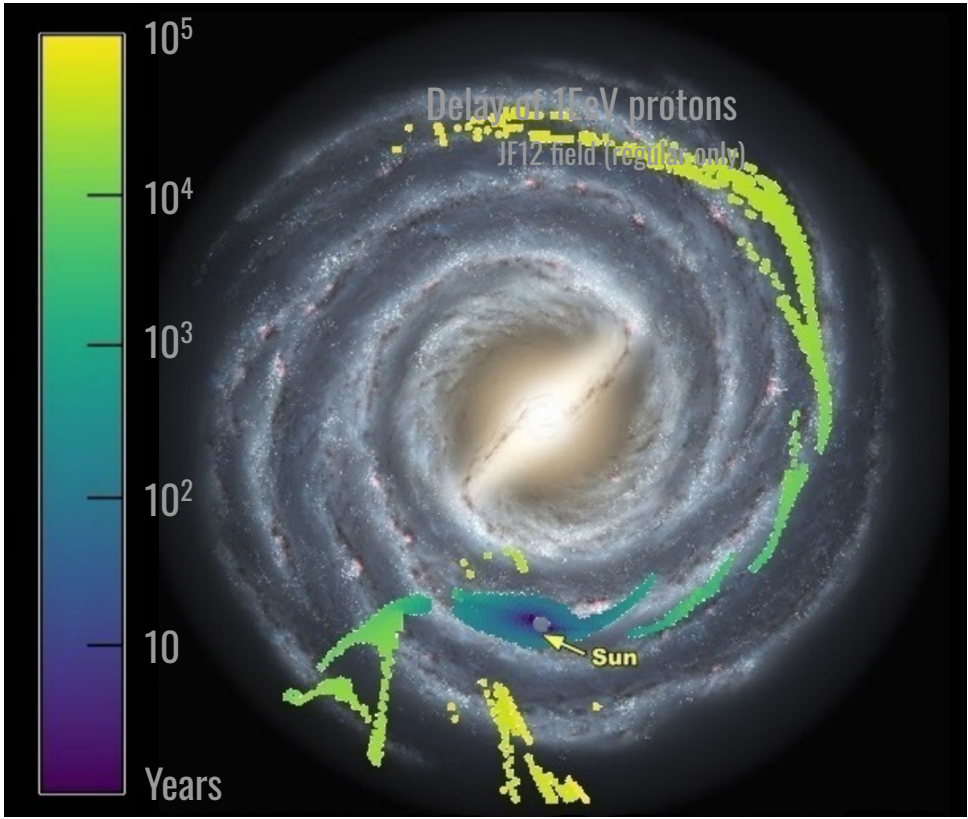


Toy model for FB emission:
-2.7 power law spectrum normalized to the excess flux observed in the direction of CenA, rescaled with distance

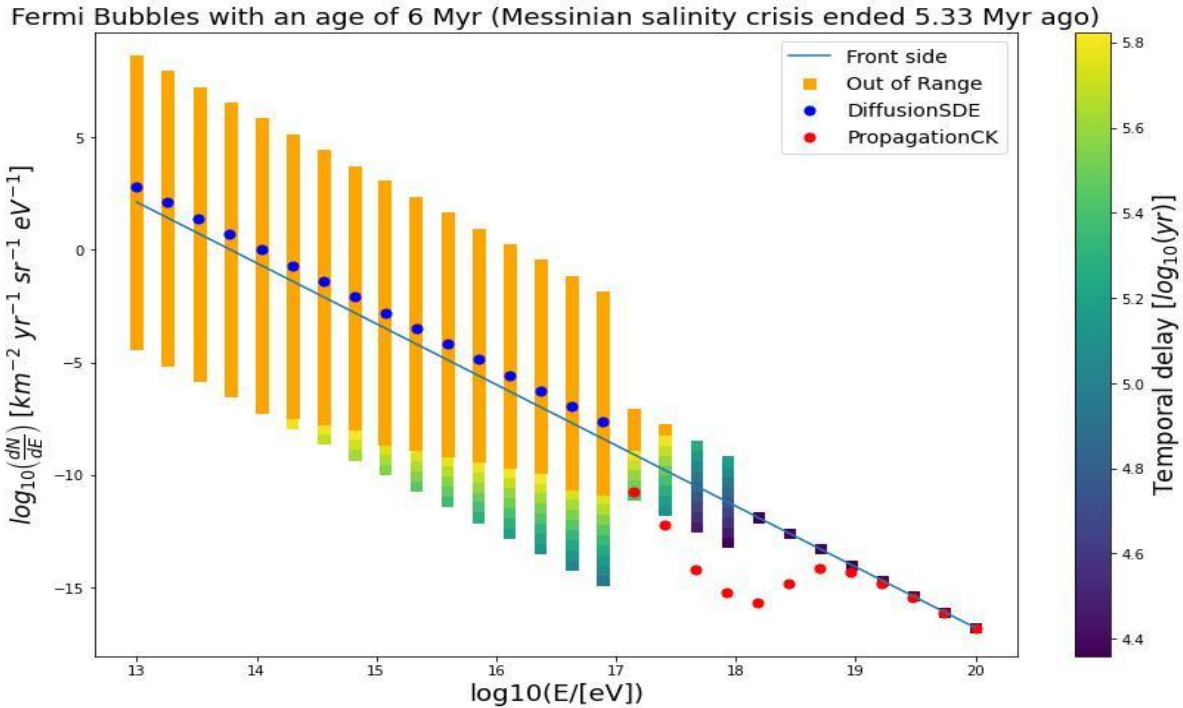
Propagation in GMF (JF12) using CRPropa (Diffusion+ballistic)

Cosmic ray delay in Galactic Magnetic Field

- Measuring the GMF (and even more extragalactic magnetic fields) is a difficult task, obtained through measurements of starlight polarization, Faraday rotation measure (RM) and synchrotron emission.
- Different models exist, which can disagree strongly on the direction of the deflection, but usually agree at least on the order of magnitude of its absolute value
- Delay is rarely considered -> need for dedicated simulations
- Below a certain rigidity (10-100 PV?) likely a transition from ballistic to diffuse propagation occur



Case study 2: MSC and the Fermi bubbles



Simulated **observed spectrum** and the **GMF induced delay** at earth

orange means out of temporal bounds

Phase 2: Simulate using Corsika the expected muon flux associated with the “observed” FB spectrum

Phase 3: Compute the nuclear recoil spectrum from muons hitting nuclei in a Halite target using Geant4

Headcount in the group

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11 institutes in Europe
10 institutes in Japan
19 institutes in the USA

Plasma cleaning halite

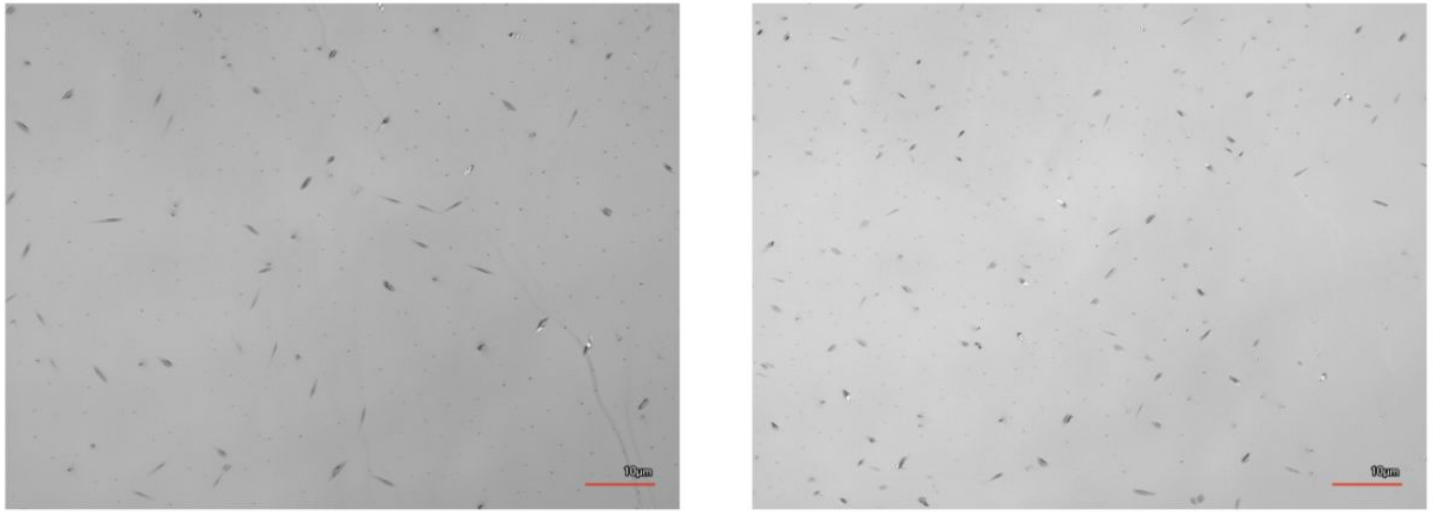


Figure 3.1: Nuclear recoil tracks revealed in Argon plasma etched halite, imaged with the Keyence VK-X1000 Laser Confocal Microscope at UNF's Materials Science and Engineering Research Facility (MSERF).

Towards a Prototype Paleo-Detector for Supernova Neutrino and DarkMatter Detection
Authors: Emilie LaVoie-Ingram, Greg Wurtz, Chris Kelso, and Zane Cable, University of North Florida+
Mineral Detection of Neutrinos and Dark Matter 2024, Proceedings



11050-AX

SPI Supplies Plasma Prep III Etcher, Solid State Design, 220 Volt

Comment

$11,578.00 each | 1 | $11,578.00

Shopping Cart Total **$11,578.00**

Product Details | Documents

SPI Supplies Plasma Prep III Etcher, with Pyrex or Quartz Chamber options.

The Plasma Prep III RF plasma-based unit is a compact tabletop design with ease of use to deliver the 1 to 100W system performance. Furthermore, the small footprint allows the system to fit neatly in most any laboratory environment.

Features:

Flexible range. Capable of 1 to 100W operation
Greater stability
Reproducible conditions.
Same small footprint as the PP II (10.5" H X 14.75" W X 12" D measurements)
Integrated vacuum gauge
Measurement and adjustment of forwarding and reverse power
Digital metering
110/220V compatible
4" diameter x 6" depth Pyrex or Quartz Chamber

Who would use this system?
The basic functions and use of the Plasma Prep III are identical to that of the long selling RF tubed based PP II system. For those users who have long process times (in terms of tens of hours to days), the solid-state unit provides for excellent stability so that the user can run the system overnight without concern about a change in operating conditions.

For those with or looking at multiple systems, the PP III allows easy adjustment of system conditions so those units in the same laboratory or even those in different locations can be run in a similar fashion. Of course, there still remain a number of external variables that may affect these systems including gas and gas pressures and the difference in line voltages.

Part Number	Item	Product URL	Qty	Price	Subtotal	Justification	Category
WFA1000	Transmitted Light Illumination / DIC Imaging Module 30 mm Cage Compatible	https://www.tho	1	4481,25	4481,25	Main transmitted light illumination module	Illumination
WFA0150	95 mm Dovetail Clamp for WFA1000 and WFA1100 Modules	https://www.tho	1	336,45	336,45	Illumination module support	Adapters
WFA1010	Warm White Illumination Kit	https://www.tho	1	814,02	814,02	Warm white (IR not necessary)	Illumination
LEDD1B	T-Cube LED Driver 1200 mA Max Drive Current (Power Supply Not Included)	https://www.tho	1	309,95	309,95	Illumination driver	Illumination
KPS201	15 V 2.66 A Power Supply Unit with 3.5 mm Jack Connector for One K- or T-Cube	https://www.tho	1	35,19	35,19	Illumination power supply	Illumination
M32M25S	Brass Microscope Adapter with External M32 x 0.75 Threads and Internal M25 x 0.75 Threads	https://www.tho	1	20,33	20,33	Adapter for objectives	Adapters
M32RMSS	Brass Microscope Adapter with External M32 x 0.75 Threads and Internal RMS Threads	https://www.tho	1	24,27	24,27	Adapter for objectives	Adapters
CEA1350	Cerna Microscope Body with Epi-Illumination Arm 350 mm Rail	https://www.tho	1	858,34	858,34	Microscope body. Sample size would not vary excessively for 350 mm rail	Body
LAURE1	Cerna Trinoculars with 10X Eyepieces Upright Image IR Filter	https://www.tho	1	3894,16	3894,16	Trinoculars and eyepieces for wide variety of structural defect size	Optics
TC1X	1X Camera Tube for LAURE1 & LAURE2 Trinoculars C-Mount Male D2N Dovetail	https://www.tho	1	384,62	384,62	Camera tube for imaging	Imaging
WFA2001	Epi-Illuminator Module for 1 Cube Conditioning Optics Male & Female D1N Dovetails	https://www.tho	1	1761,23	1761,23	Light guide module for even light across the field of view	Illumination
CSC1001	Nikon FN-C LWD Condenser 0.78 NA Male D3N Dovetail	https://www.tho	1	2354,15	2354,15	Main condenser	Illumination
CSN100	Nosepiece for 1 Objective M32 x 0.75 Threads 60 mm Cage Compatible	https://www.tho	1	104,67	104,67	Support fo objective	Adapters
CSA2000	Condenser Arm ±2 mm Travel in X & Y Female D3N Dovetail 60 mm Cage Compatible	https://www.tho	1	717,35	717,35	Support for condenser	Adapters
PLSZ	Motorized Module with 1" Travel 95 mm Dovetail	https://www.tho	2	1688,56	3377,12	Auto-focusing module	Focusing
PLSZ1	Angle Bracket for Edge-Mounted Arms	https://www.tho	2	161,44	322,88	Bracket for focusing module	Adapters
MC301	Three-Channel Controller for Motorized Rigid Stands and PLS Series Stages	https://www.thor	1	1437,13	1437,13	Controller for autofocusing module	Focusing
MCMK3	3-Knob USB HID Joystick	https://www.tho	1	1031,26	1031,26	Joystick for autofocusing module	Focusing
LPXY1	Low-Profile Motorized XY Scanning Stage 110 mm x 75 mm Travel	https://www.tho	1	4357,75	4357,75	Motorized stage for quick automated scanning fo the samples	Stage
CSA1000	Fixed Arm Holder for MLS203-1 Fast XY Scanning Stage	https://www.tho	1	706,33	706,33	Support for motorized stage	Adapters
BSC202	Two-Channel Benchtop Stepper Motor Controller	https://www.tho	1	2333,93	2333,93	Controller for motorized stage	Stage
MJC2	2-Axis USB HID Joystick	https://www.tho	1	489,55	489,55	Joystick for motorized stage	Stage
					30151,93		

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Item / Description	Price / Availability	Quantity	Subtotal
 LEDD1B T-Cube LED Driver, 1200 mA Max Drive Current (Power Supply Not Included)	~~322,86 €~~ * 309,95 € Today	1 Remove	309,95 €
 KPS201 15 V, 2.66 A Power Supply Unit with 3.5 mm Jack Connector for One K- or T-Cube	~~36,66 €~~ * 35,19 € Today	1 Remove	35,19 €
 M32M25S Brass Microscope Adapter with External M32 x 0.75 Threads and Internal M25 x 0.75 Threads	~~21,18 €~~ * 20,33 € Today	1 Remove	20,33 €
 M32RMSS Brass Microscope Adapter with External M32 x 0.75 Threads and Internal RMS Threads	~~25,28 €~~ * 24,27 € Today	1 Remove	24,27 €

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 CSC1001 Nikon FN-C LWD Condenser, 0.78 NA, Male D3N Dovetail	~~2.452,24 €~~ * 2.354,15 € 7-10 Days	1 Remove	2.354,15 €
 CSN100 Nosepiece for 1 Objective, M32 x 0.75 Threads, 60 mm Cage Compatible	~~109,03 €~~ * 104,67 € Today	1 Remove	104,67 €
 CSA2000 Condenser Arm, ±2 mm Travel in X & Y, Female D3N Dovetail, 60 mm Cage Compatible	~~747,24 €~~ * 717,35 € Today	1 Remove	717,35 €
 PLSZ Motorized Module with 1" Travel, 95 mm Dovetail	~~1.758,91 €~~ * 1.688,56 € Today	2 Remove	3.377,12 €

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Item / Description	Price / Availability	Quantity	Subtotal
 CEA1350 Cerna Microscope Body with Epi-Illumination Arm, 350 mm Rail	~~894,10 €~~ * 858,34 € Today	1 Remove	858,34 €
 LAURE1 Cerna Trinoculars with 10X Eyepieces, Upright Image, IR Filter	3.894,16 € Today	1 Remove	3.894,16 €
 TC1X 1X Camera Tube for LAURE1 & LAURE2 Trinoculars, C-Mount, Male D2N Dovetail	~~400,65 €~~ * 384,62 € 7-10 Days	1 Remove	384,62 €
 WFA2001 Epi-Illuminator Module for 1 Cube, Conditioning Optics, Male & Female D1N Dovetails	~~1.834,61 €~~ * 1.761,23 € Today	1 Remove	1.761,23 €

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Item(s)	22
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Item / Description	Price / Availability	Quantity	Subtotal
 PLSZ1 Angle Bracket for Edge-Mounted Arms	~~168,16 €~~ * 161,44 € Today	2 Remove	322,88 €
 MCMK3 3-Knob USB HID Joystick	~~1.074,23 €~~ * 1.031,26 € Today	1 Remove	1.031,26 €
 LPXY1 Low-Profile Motorized XY Scanning Stage, 110 mm x 75 mm Travel The quantity you have requested for part number LPXY1 exceeds available stock. After you have finalized your order, a customer representative will contact you with an exact delivery date, or you are welcome to contact us by phone at +49 (0) 8131-5956-0 or by email at sales.de@thorlabs.com	4.357,75 € Lead Time	1 Remove	4.357,75 €

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Item / Description	Price / Availability	Quantity	Subtotal
 CSA1000 Fixed Arm Holder for MLS203-1 Fast XY Scanning Stage	~~735,76 €~~ * 706,33 € 7-10 Days	1 Remove	706,33 €
 BSC202 Two-Channel Benchtop Stepper Motor Controller	~~2.431,18 €~~ * 2.333,93 € Today	1 Remove	2.333,93 €
 MJC2 2-Axis USB HID Joystick	~~509,95 €~~ * 489,55 € Today	1 Remove	489,55 €
 MCM301 Three-Channel Controller for Motorized Rigid Stands and PLS Series Stages	~~1.437,13 €~~ * 1.379,64 € Today	1 Remove	1.379,64 €

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Subtotal	30.094,44 €
Package Weight	46.15 kgs

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